Class 8: Analysis of Breast Cancer Cells

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Background

In today's class we will be employing all the R techniques for data analysis that we have learned thus far - including the machine learning methods of clustering and PCA - to analyze real breast cancer biopsy data.

Data Import

The data is in CSV format:

```
wisc.df <- read.csv("WisconsinCancer.csv", row.names = 1)
```

wee peak at the data

```
head(wisc.df, 3)
```

```
         diagnosis radius_mean texture_mean perimeter_mean area_mean
842302           M       17.99        10.38          122.8      1001
842517           M       20.57        17.77          132.9      1326
84300903         M       19.69        21.25          130.0      1203
         smoothness_mean compactness_mean concavity_mean concave.points_mean
842302           0.11840          0.27760         0.3001             0.14710
842517           0.08474          0.07864         0.0869             0.07017
84300903         0.10960          0.15990         0.1974             0.12790
         symmetry_mean fractal_dimension_mean radius_se texture_se perimeter_se
842302          0.2419                0.07871    1.0950     0.9053        8.589
842517          0.1812                0.05667    0.5435     0.7339        3.398
84300903        0.2069                0.05999    0.7456     0.7869        4.585
         area_se smoothness_se compactness_se concavity_se concave.points_se
842302    153.40      0.006399        0.04904      0.05373           0.01587
842517     74.08      0.005225        0.01308      0.01860           0.01340
84300903   94.03      0.006150        0.04006      0.03832           0.02058
         symmetry_se fractal_dimension_se radius_worst texture_worst
842302       0.03003             0.006193        25.38         17.33
842517       0.01389             0.003532        24.99         23.41
84300903     0.02250             0.004571        23.57         25.53
         perimeter_worst area_worst smoothness_worst compactness_worst
842302             184.6       2019           0.1622            0.6656
842517             158.8       1956           0.1238            0.1866
84300903           152.5       1709           0.1444            0.4245
         concavity_worst concave.points_worst symmetry_worst
842302            0.7119               0.2654         0.4601
842517            0.2416               0.1860         0.2750
84300903          0.4504               0.2430         0.3613
         fractal_dimension_worst
842302                   0.11890
842517                   0.08902
84300903                 0.08758
```

```r
nrow(wisc.df)
```

```
[1] 569
```

Q1. How many observations are in this dataset?

There are 569 observations in the dataset.

Q2. How many of the observations have a malignant diagnosis?

```r
table(wisc.df$diagnosis == "M")
```

```
FALSE  TRUE 
  357   212 
```

There are 212 individuals who are malignant in their biopsy sites.

> Q3. How many variables/features in the data are suffixed with _mean?

There are 10 variables that are suffixed with _mean.

```r
length(grep("_mean", colnames(wisc.df)))
```

```
[1] 10
```

We need to remove the `diagnosis` column before we do any further analysis of this dataset - we don't want to pass this to PCA etc. We will save it as a separate wee vector that we can use later to compare our findings to those of experts.

```r
wisc.data <- wisc.df[,-1]
diagnosis <- wisc.df$diagnosis
```

Principal Component Analysis

The main function in base R is called `prcomp()`we will use the optional argument `scale=TRUE` here as the data columns/features/dimensions are on very different scales in the original data set.

```r
wisc.pr <- prcomp(wisc.data, scale=T)
```

```r
attributes(wisc.pr)
```

```
$names
[1] "sdev"     "rotation" "center"   "scale"    "x"

$class
[1] "prcomp"
```

```r
wisc.pr$x
```

	PC1	PC2	PC3	PC4	PC5
842302	-9.18475521	-1.946870030	-1.122178766	3.630536408	1.1940594778
842517	-2.38570263	3.764859063	-0.528827374	1.117280773	-0.6212283647
84300903	-5.72885549	1.074228589	-0.551262540	0.911280840	0.1769302185
84348301	-7.11669126	-10.266555635	-3.229947535	0.152412923	2.9582754313
84358402	-3.93184247	1.946358977	1.388544953	2.938054169	-0.5462667446
843786	-2.37815462	-3.946456430	-2.932296681	0.940209586	1.0551135428
844359	-2.23691506	2.687666414	-1.638471247	0.149208600	-0.0403240371
84458202	-2.14141428	-2.338186649	-0.871180715	-0.126931171	1.4261817833
844981	-3.17213315	-3.388831138	-3.117243065	-0.600768440	1.5209521144
84501001	-6.34616284	-7.720380945	-4.338098744	-3.372234369	-1.7087596141
845636	0.80970132	2.656937673	-0.488400053	-1.671096176	-0.2755690974
84610002	-2.64876984	-0.066509405	-1.525113359	0.051216497	-0.3316592904
846226	-8.17783882	-2.698602007	5.725193241	-1.111278747	-1.0425531111
846381	-0.34182514	0.967428026	1.715661982	-0.594479872	-0.4675990720
84667401	-4.33856172	-4.856809832	-2.813639832	-1.453278303	-1.2889287269
84799002	-4.07207318	-2.974443983	-3.122526655	-2.455909915	0.4079831402
848406	-0.22985277	1.563382114	-0.801813642	-0.650010973	0.4942761390
84862001	-4.41412695	-1.417423150	-2.268323145	-0.186108659	1.4226094459
849014	-4.94435304	4.110716528	-0.314472408	-0.088128974	0.0566653163
8510426	1.23597583	0.188049490	-0.592761930	1.594942716	0.4417655273
8510653	1.57677384	-0.572304625	-1.799863016	1.124286473	0.3949222435
8510824	3.55420904	-1.661487969	0.450790799	2.071941593	0.4903150736
8511133	-4.72904972	-3.302058265	-1.465247427	2.039355666	0.0261970715
851509	-4.20482441	5.123858059	-0.751740573	-0.861951843	0.4705538779
852552	-4.94528075	1.542395143	-1.711687827	0.046718059	1.7377012054
852631	-7.09232236	-2.016835734	-0.028984132	2.585675482	2.0375470912
852763	-3.50717666	-2.169715994	-3.891122637	-1.294620549	0.0681580703
852781	-3.06136021	1.874902631	2.579478032	0.128371462	0.1165625639
852973	-4.00374127	-0.536769860	-2.759197831	-1.896718595	0.5252107652
853201	-1.71380176	1.522365502	0.146058578	1.909706071	-0.5356519265
853401	-6.05411853	0.756511800	-0.348932376	-1.063281958	0.7454493340
853612	-2.89968469	-4.001774373	-2.998823809	-0.325030387	0.8546038749
85382601	-4.55077848	-0.337239419	-0.753637681	-0.699714306	1.5471839770
854002	-4.98621538	1.131593223	-2.575118337	-1.216191663	-0.6693259394
854039	-2.98271631	-0.757756497	-3.490690279	0.054212589	-0.1669530505
854253	-2.76393718	0.354044421	-1.895295337	-0.533786198	0.8832432812
854268	-1.29505925	-0.912393466	-1.574383382	-1.042249570	-0.4143571251
854941	3.74601730	1.412230504	1.730784885	-0.564945456	-0.0802687991
855133	0.99719148	3.348346731	4.301788720	-0.560716245	0.4639645611

855138	-0.76459136	-0.885464837	-2.702540362	-0.274538185	-0.8097169787
855167	2.14906252	1.922300194	-1.179446370	-1.112219391	-0.4480774771
855563	0.09324934	-2.258764532	-1.903051997	-1.441039161	1.7038647386
855625	-9.08001023	-2.016898441	2.139343894	-1.068304028	-0.7972299505
856106	-0.98958304	-0.984064148	-2.308702241	-0.276582144	0.6399478981
85638502	0.29328849	0.136978564	-3.299029123	-1.008898141	0.0224910920
857010	-5.37620991	0.134758404	-1.677366657	1.506369463	-0.1518645822
85713702	4.57790860	-1.482915508	0.042037623	0.295968708	0.7132249498
85715	-1.69851237	-2.350203862	-3.075383081	-0.065962912	1.0555376469
857155	2.13456708	-0.095745364	-1.491099878	1.120077655	0.0679774156
857156	1.56610099	1.207370855	-0.368834570	-1.474331566	-0.1725985690
857343	3.53979091	1.281368110	1.017315805	-0.607621596	0.2194428358
857373	3.15503795	1.687473796	-0.495857416	0.193841543	-1.4386516886
857374	3.44745515	0.497780720	-0.152383875	0.574980496	0.0522688899
857392	-3.29964761	1.129943574	0.825666100	0.981853085	1.1942122234
857438	0.67402621	2.114549137	-1.394992302	-1.042221635	-0.2416042153
85759902	2.85564577	-0.152588874	0.428557855	0.482635650	1.5092086632
857637	-4.64465207	2.308301508	-0.728168145	0.639379104	1.1573061797
857793	-2.17494916	-0.971261438	-1.942179421	-0.736670095	0.3954358798
857810	3.71818738	1.786070097	1.278807604	0.115464169	0.5258196576
858477	4.13232693	-2.401679261	-1.241865694	2.046545722	1.1646276518
858970	2.38373822	-2.755233814	1.213018303	1.792911568	3.3352233901
858981	2.57661610	-3.135912650	1.399049169	-0.763194998	2.7816479152
858986	-4.75492832	-3.009032875	-0.166094576	-0.605295233	-1.1406976712
859196	2.31209785	-3.265116721	2.793766025	0.565229094	0.3814984103
85922302	-1.69012080	-1.539322116	-1.798262954	-1.389367072	1.2116088977
859283	-1.81071217	-0.722104815	-1.464960560	-1.197676214	1.4267861675
859464	2.78347559	-2.308617372	0.472309720	-1.605747397	1.6114402596
859465	3.51555502	0.657730736	0.590855071	-0.010793189	-0.0205709988
859471	-4.32619605	-9.194435552	1.491650337	0.319666023	-4.7240139158
859487	3.25841241	0.937013647	0.204949679	1.089113223	0.4599452518
859575	-2.70221851	4.433240985	0.307075192	0.488263229	-0.3721675732
859711	0.30758513	-7.381317396	3.815728542	2.131680166	-2.7083196796
859717	-5.49886689	-0.937500513	-2.208188675	-1.522456429	-0.9240784393
859983	0.36139121	-0.119633821	-2.039161544	1.057953093	-0.7564437850
8610175	2.62766457	0.696696349	-0.583428281	0.337884296	-0.4671344130
8610404	-1.42691206	1.965372092	1.110961327	0.752248984	0.5394071070
8610629	0.83378424	-1.963876860	0.789737653	3.434609101	2.3524809639
8610637	-6.22541880	-0.919260690	0.089748341	2.536926544	-0.0620812262
8610862	-11.65845644	-4.744442593	2.004115633	0.318768543	3.3883795562
8610908	2.01980045	0.254675746	-0.647302452	0.059199217	0.0547996477
861103	1.63694460	-1.714440587	0.433824539	-1.526622964	1.7678261507
8611161	-1.16643527	-2.512305286	-1.904043397	0.637750373	-0.2359734956

8611555	-10.75977535	2.255997858	0.038801648	-0.389043228	-1.1862447821
8611792	-5.03038488	-0.773728357	2.679979922	-1.665039071	0.3113456414
8612080	2.17255269	-0.496441169	-0.948257444	0.207037052	0.8845677371
8612399	-3.28534463	1.666770480	0.146802675	0.437108401	1.3673667927
86135501	-0.60707378	-0.162071919	1.635935419	-1.113531850	0.5403198078
86135502	-3.58041324	2.204721916	-1.703037100	-0.800589442	-0.3650009185
861597	0.93333687	-0.926885919	-0.040128345	-1.352209082	-0.0648850467
861598	-1.25849730	-1.014684255	-0.080521995	2.049531496	-0.1373994614
861648	1.58686770	1.618232954	0.309726682	-1.091038889	-0.0537268234
861799	-0.25227559	0.530884425	0.588809736	-0.110441165	-0.4283042881
861853	2.84492835	2.891103732	0.527524677	0.598290300	-1.4344348070
862009	1.96322268	0.964308498	-0.081136715	-0.093998029	0.5495403920
862028	-2.77342525	-0.557509977	-0.605238702	0.499390598	-0.4994662006
86208	-4.39236314	2.121640892	0.579000549	-0.776729960	0.3035669597
86211	2.58980407	-0.213446120	1.034701089	0.661423462	0.5402889624
862261	3.90090576	-1.189020421	1.550616806	-0.934952821	0.7074978792
862485	2.81575365	-0.367560567	-1.350247556	1.736886645	-0.2840715251
862548	-0.61573908	-0.638349988	-0.199743121	-1.096533937	-0.1185853914
862717	0.43247968	1.390820523	-0.082646762	-1.933578565	0.0344302397
862722	4.55102442	-3.525683788	-0.215074716	1.090925794	2.7559240651
862965	3.44515217	1.423370040	-0.088646005	-1.709223842	0.1065229834
862980	2.11410008	-1.847747782	-0.556184595	-0.463948150	0.4744968363
862989	2.68898618	-1.418813334	0.968298772	-0.002857011	1.2554137475
863030	-3.21109134	-4.043198634	-2.712997305	1.327457319	0.2214783192
863031	0.74861266	-1.796058175	-0.567866143	-0.560006237	0.9322757208
863270	3.15622889	1.034838878	-1.489373670	-0.592646911	-0.5136339075
86355	-13.04464395	-0.980650357	0.598286247	1.471636087	0.5993856696
864018	2.22672053	-0.666823902	-1.090600727	-1.077353762	0.2988661489
864033	2.49063954	-2.596401708	1.668257671	0.632929090	1.1557305724
86408	0.10325957	-2.278139015	2.107708229	-0.351536677	-0.6219347413
86409	-3.58813694	-3.922881433	2.045207163	-0.231562584	-5.4604591892
864292	1.34776279	-3.553098748	1.730668471	-0.046949434	0.3449474850
864496	2.50791098	-3.248461200	-1.686140767	1.188915291	0.4551865326
864685	2.04423086	-0.304616836	-0.533369397	-0.489411120	0.0678524055
864726	2.09522559	-3.663872062	3.629007335	1.538433194	-2.2345636748
864729	-3.10779316	-1.568009873	-2.611638159	0.617924024	1.1807403791
864877	-4.95236801	-2.382749650	-1.933405900	-0.610304229	0.1480794337
865128	-0.85026612	2.304707468	0.566963050	-0.043586897	2.0922055990
865137	2.96339173	-0.371179883	-2.064716427	2.070684315	-0.1628943517
86517	-3.33120209	1.324392029	0.727373652	1.002316780	0.9328861016
865423	-12.88327621	-2.314585873	6.323133317	1.940496559	3.1574494112
865432	0.77006610	0.064052860	-1.133411514	2.508256103	-0.0202192366
865468	2.20057925	0.734958544	0.266845303	-0.037586447	-2.9314571580

86561	3.14064881	1.875758557	-0.669779705	0.303065257	-0.4077106627
866083	0.63831885	0.910564828	-2.140840449	-2.139033534	0.4287624850
866203	-1.91744627	3.534971966	1.253821297	0.335703696	-0.5885511025
866458	-1.40762980	-1.303782215	0.640864876	1.885297508	0.0362201276
866674	-4.63960895	1.480714297	-1.169651383	-1.217534223	-0.1779160105
866714	1.87581684	-1.421979452	-1.069870821	1.707830258	0.8837714798
8670	-1.43081046	1.048681220	-1.260901861	0.532720319	0.6261456617
86730502	-1.35143790	1.153126842	-1.208729712	-0.827378497	0.9859398192
867387	0.70849143	1.566853657	-0.669405084	1.685780647	-0.3390165106
867739	-2.16950708	2.823776101	-0.561613332	-0.660250673	0.4745525764
868202	1.97510247	0.419018755	-0.380240597	-1.737931013	0.3083668329
868223	2.59850310	0.481911436	1.870726668	-0.437633248	-0.0160686443
868682	3.25514318	0.417953651	-0.691197447	0.659273518	-0.1214593563
868826	-3.77870160	-0.859625218	3.667773435	2.046019895	2.1818702714
868871	1.99028348	-1.328186508	1.134246527	2.044634146	0.2519061287
868999	5.01059989	-0.574194748	-0.841697543	2.167755281	0.8812268193
869104	-1.34383800	1.273650278	0.555354894	0.574366142	0.4704353120
869218	2.45380972	-0.897665441	0.352823033	-0.367668079	1.1033160270
869224	1.83566648	0.090946742	-1.444010288	0.671957291	0.1360791853
869254	4.34266862	0.892786708	0.638694049	0.425252734	-0.9556015217
869476	0.73216868	-3.698927716	0.657926113	2.234161633	0.3361716375
869691	-2.39788795	-4.833735642	-1.027253164	0.342615376	2.3897699977
86973701	-0.39275308	-1.082115915	2.245629588	-0.461452569	-1.6901413981
86973702	0.41196533	0.389288784	-1.044851048	1.365044596	-0.8627479982
869931	3.04724369	2.235817239	-0.469040276	0.182614048	-1.5744606816
871001501	1.44129554	-0.305601552	0.551062974	0.355761009	2.7005428427
871001502	-0.08311297	-7.144073768	-0.059957215	-1.801496676	-0.7374637239
8710441	-7.08707084	-12.562140869	7.352238887	1.224264329	-7.4838486945
87106	3.74011364	-0.250281306	-0.097297641	1.930448497	1.0426741051
8711002	0.96832079	-0.944113582	-0.890028129	1.095568323	0.6129630899
8711003	2.41659356	-0.005547532	-0.797678511	-0.070829748	0.0772419138
8711202	-4.09718264	0.378470806	1.445046044	0.701751100	-0.0006644087
8711216	0.75094226	3.067949187	1.447907672	-0.952587697	-0.8911446262
871122	3.65143365	0.674055644	-0.906188637	1.509832817	-0.2935538042
871149	4.67609710	1.102886823	-0.257081973	1.245379911	-0.4142538767
8711561	0.59725624	-1.784081072	1.480433343	-0.177715866	1.5878294278
8711803	-3.38435289	2.908477953	3.004594027	2.825801426	-0.5078995524
871201	-6.14447970	2.015878994	-1.564766888	0.970791871	0.3360742118
8712064	1.32386648	-1.468002337	0.847675513	-1.141761837	0.0477797028
8712289	-5.48932277	4.162167037	-0.600732185	0.149568094	-0.6250164993
8712291	2.99476813	2.736453261	-0.161042080	-0.412806573	-0.4051896121
87127	4.38287625	-0.006866240	-0.947274735	2.795545994	-0.4453390247
8712729	-1.21256768	2.037246395	0.979834026	0.238756035	-0.0856932721

8712766	-5.06520424	1.783582969	1.172615291	-0.693122956	-0.6480853131
8712853	1.97596280	1.841159394	-0.277115927	0.549266608	0.2067144089
87139402	2.51266554	-0.114151857	-0.496186180	2.352277364	0.6887689035
87163	0.94665494	1.683272444	-0.350877607	-0.576703303	0.3412418770
87164	-2.81688978	-1.263990953	-2.151251366	2.706077979	0.0400092558
871641	3.48092345	-1.618268077	2.673117328	1.252511728	0.5984548400
871642	4.65463399	0.222718846	1.557987797	0.510643007	1.8952731945
872113	5.34691339	-1.025855214	0.324700222	1.367479190	0.9178537547
872608	-1.16986842	-7.008319996	4.268106508	-0.332347396	-2.9068870248
87281702	-2.95370292	-0.705800822	-1.041419103	-0.521690540	-1.4123928506
873357	4.97132782	3.383227983	0.439451738	-1.576183104	-1.0328868993
873586	4.06045289	1.245070643	0.573972801	1.703893823	-0.6153230400
873592	-9.50430706	5.598558052	-0.636679193	0.224024081	0.4061861282
873593	-8.99924714	-0.580520281	-3.131959404	-0.861511555	-0.0456818494
873701	-0.75821134	1.607118481	-1.513763762	-1.075114038	0.0491913464
873843	2.65702172	-0.539461132	1.705514033	1.200694830	-2.3526717446
873885	0.38967611	0.988372689	-2.582896564	-0.783837500	-0.7336657053
874158	3.88564657	-0.815354285	1.014824328	0.824193256	2.1316971080
874217	-0.36455538	3.571318842	-2.222023387	0.222971846	-0.7778465452
874373	2.83339783	0.398379162	-0.347139642	0.812532578	0.0392461561
874662	3.30737348	-0.155604280	0.541002917	-0.512033374	1.7863073596
874839	3.35435293	1.102762109	0.151956099	0.827459950	-0.7336137377
874858	-6.51738201	-8.004126824	-0.326136731	-2.769361777	-1.5975261925
875093	1.71622508	0.542688019	3.295475582	-0.087253972	-0.0840995519
875099	5.56084294	0.477427493	4.123834207	-2.350422636	1.0515357688
875263	-1.77809695	-2.774146213	-0.941934046	-3.071649469	0.6945219915
87556202	-2.60918908	-1.560049030	-0.217995141	-0.492342056	-1.1034164923
875878	2.81656023	0.969257722	-0.382255094	0.463382873	-0.5513017565
875938	-2.49624245	-2.276479790	1.318318594	-1.179800454	1.8980601601
877159	-1.27590451	2.441111044	2.464148795	-0.129776372	-2.1534113490
877486	-3.47014398	2.275846405	-1.237595112	-1.112162383	-1.0083717623
877500	-1.25557034	-0.382057094	-3.834946034	-0.716289870	0.6505837048
877501	1.47213604	-0.116787496	-0.214479145	-0.490377374	0.7132142476
877989	-1.64801062	2.100442902	-0.974414210	0.187878371	-0.9160201700
878796	-9.02864522	0.654596849	0.221085545	-0.764058327	-0.5644000271
87880	-4.55058603	-3.083925626	-2.268408346	-2.189192919	3.4583122917
87930	0.78050321	-0.652275325	-0.643091145	0.214279700	0.4372951306
879523	0.22289648	0.701204504	-2.268765765	0.997753580	-0.6459195774
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8810158	-0.31416180	-2.075734301	-2.081480386	-1.067600421	0.2176779819
8810436	2.05738166	2.470613908	-1.460433790	1.866439637	-1.8369468233
881046502	-4.80474158	3.026440019	2.741606351	0.283114502	-0.1937940455

8810528	2.99607430	0.396429212	-0.595941573	-0.207862040	-0.5678601226
8810703	-12.27421974	7.536778599	10.103533703	5.189703614	3.8491201120
881094802	-3.36923202	-2.585550416	7.085840019	-0.367171665	-2.7681374358
8810955	-2.50656080	-2.612349885	0.206198902	-1.893080135	2.1228977028
8810987	-1.31690973	-2.152585453	-1.785061911	-0.010160725	-0.1670054272
8811523	0.38841453	-2.274796314	0.626780574	-0.429349217	-0.1192752256
8811779	2.75450931	-1.085879226	1.705737291	-0.516377016	-0.8063132292
8811842	-4.93923501	2.845820436	0.913075404	0.334889341	0.4829010175
88119002	-4.29429880	4.662172958	-0.006834994	-3.650173128	-0.1785458275
8812816	2.37141521	0.732757814	-1.524685925	1.851700285	-1.1820779548
8812818	0.96634003	-0.438052741	-1.446423567	1.988611655	0.0938834761
8812844	2.97318270	-1.809381711	-0.223238842	0.391290904	1.5280262811
8812877	-1.80166175	-0.166314618	-2.626288708	-0.643175553	0.5214235661
8813129	2.47618073	1.417329346	-0.316254972	0.328213945	-0.6614306092
88143502	0.82531310	1.249148354	0.267725157	2.308595141	1.0139214702
88147101	3.89127342	-0.538097276	-0.880783727	1.037886765	0.8083176466
88147102	0.38923560	0.613875324	-1.351679911	1.174159176	-1.7883197584
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881861	-2.75314552	-3.462726769	-2.471875866	-1.007392449	-0.1549996143
881972	-3.25439115	0.125090426	-2.893551435	0.748931137	0.3525453547
88199202	4.01560693	1.353056834	-0.360155815	-2.633480577	-0.7345986421
88203002	3.67318128	1.290493562	0.424237023	-4.597664394	0.7402578603
88206102	-3.31340866	3.935689268	-0.205273062	-2.116361469	-0.9691280654
882488	4.03305870	-1.161524089	0.189581014	0.939690169	0.5651016963
88249602	2.33218128	1.347793816	0.232482459	-1.211847619	-0.0515295475
88299702	-8.39645911	4.150252991	0.046927784	-0.551357625	0.2878270868
883263	-2.64524110	3.947699392	1.370675284	0.207636553	-1.4038189712
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883539	4.47364832	1.739598699	-0.887426246	0.763980319	-0.6046080177
883852	-1.32145867	-4.785266054	-0.661952674	-1.020298051	-2.1481343740
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884626	-1.02499890	-2.359186109	-1.615464155	1.144955805	-4.1408350383
88466802	2.58028546	-0.728572047	-0.547660903	-2.330028998	1.8336515859
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888264	0.83947101	3.497008034	0.182323228	-1.241480335	0.1084218681
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8910499	2.01038350	1.102516460	1.336847795	-1.614259260	-0.6911126965
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8912280	-2.63493731	-0.576647607	-1.992946242	0.272309546	-1.1957159307
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891936	4.76551701	0.542484530	0.893082182	1.729539559	-1.0008319932

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894047	3.76210430	-4.394333943	3.946735131	-0.306998155	4.7854456445
894089	4.86695697	2.337158334	-0.168146877	0.694278950	-0.8202171408
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894604	1.29372783	-3.467757378	1.108552569	0.766005357	-0.2103251163
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89827	2.07535283	-1.804931340	-1.166053416	1.141225718	0.4132078419
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89869	1.20553704	1.315886497	-0.929922572	1.703291193	-0.7180879050
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899667	-6.17684877	-5.103514894	0.520682438	0.939912167	0.6059916654
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9010259	-0.35757923	-2.125968004	0.707586438	0.184582384	0.1024797906
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9010333	2.99060189	-1.629958015	2.554470244	1.344703963	-0.0407045185
901034301	3.19110631	-0.578830133	0.489333658	0.044554975	0.7229711106
901034302	4.58792496	2.758699732	0.979680613	0.362519486	-0.2198878361
901041	2.48373073	1.187894183	0.348470172	-1.255187427	0.1292803966
9010598	2.39655663	0.250524355	-0.326850211	-0.277129058	0.3545026443
9010872	0.77892982	2.122475872	0.320506876	0.079179253	0.0857346246
9010877	3.13722800	1.486714010	-1.197824530	0.453050754	-0.8401627046
901088	-2.73260737	3.941699950	-0.017154393	0.299846924	-0.2830181364
9011494	-6.22155168	1.388887616	2.731833465	-1.092200460	0.5821459417
9011495	2.22182078	0.356846723	-1.100713817	0.186058986	0.2390645934
9011971	-5.30247440	6.717504288	1.484358772	1.774948969	1.0726100091
9012000	-7.24162919	3.652255530	-0.155565119	1.225750236	-0.2973284171
9012315	-4.20352196	-1.175208822	-1.588620367	-0.977982825	0.1174778025
9012568	2.49743889	2.016761090	-0.730631718	1.983278202	-1.1349569236
9012795	-3.63181933	1.954720962	-0.828046542	1.378539602	-0.9935467144
901288	-3.51615443	3.855214937	-1.021630135	1.382926471	-0.3203822623
9013005	2.61264945	1.101505251	-1.239645246	0.908676936	-0.3781480185
901303	0.09602363	0.129733431	-1.669994689	1.526133855	-0.4027131208
901315	-2.04305926	-6.421169887	1.273001142	-0.109645013	-4.5986701548
9013579	3.06405348	2.180177374	-0.278817341	-2.935974046	-0.4728184934
9013594	1.60217670	-0.292338995	-1.584830948	0.908898609	-1.2347827015
9013838	-3.76262144	-5.980033549	-4.876070721	-1.119887331	0.5637410178
901549	0.92475292	-2.300457270	-1.594708683	1.674764766	1.4257318574
901836	3.28545874	-0.201134359	-0.627204410	0.612295926	-0.1706180435
90250	1.55348525	-0.978996316	1.195507876	-1.773616381	-3.4960491469

90251	0.39842934	-2.159312724	-0.944773717	0.372153562	-1.0117604959
902727	2.29854534	0.931258175	-1.128959899	1.451099756	-1.5542727004
90291	0.54242409	1.315910048	0.994879134	-1.483570015	-0.2027995563
902975	2.05700950	-0.320153445	-0.122350756	1.019962435	-2.0773397982
902976	3.42150749	2.443368505	0.045009146	0.785450794	-1.6017729103
903011	0.47199125	-3.699574203	1.952067274	0.995440694	-3.8110426797
90312	-4.69936579	0.195783486	3.753730061	-1.003344863	-0.3283619127
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903483	3.59152918	-2.602797030	1.850401478	0.089806410	2.4403924824
903507	-4.14544978	-0.766814129	-0.887194062	-0.008362190	0.6044479220
903516	-7.66539681	0.859727681	-2.151756805	0.254499442	0.2362625219
903554	2.00861205	-0.429311811	-0.103997306	-0.160151144	1.5081790902
903811	2.91826573	1.698977554	0.295467269	-0.413812779	-0.3635999269
90401601	0.50029759	-0.106637635	-1.027843564	-0.304616687	-0.1389987468
90401602	1.90981814	0.650821226	1.449013512	0.400350786	-2.0611887662
904302	3.78177167	0.325859645	-1.057150017	0.605475000	-1.5248138903
904357	2.70601296	0.219472629	-0.314228262	0.055204339	0.0482116178
90439701	-6.28414409	-2.034757025	-3.179284831	0.155406788	-0.2023110488
904647	3.42775463	0.991737410	-0.731051348	1.342541631	-0.2356318119
904689	1.99962900	0.293068594	0.618659534	-0.352201845	-1.1637958353
9047	2.47111877	0.334752480	-1.481058853	0.464828450	-0.0489496329
904969	3.64405137	1.240273815	1.385405524	1.612540488	0.1728393129
904971	2.34412427	-0.682141105	1.131738080	-0.302837266	0.5411994776
905189	1.14199209	1.960446508	-1.239628958	1.477776286	-0.4837568556
905190	1.34966974	-0.369249002	2.484489568	-1.139714833	-1.9041890533
90524101	-2.76094145	1.077685260	-1.178586912	0.413794976	0.1644668356
905501	1.96759233	0.175693916	0.420361503	-0.785620740	0.8604953761
905502	2.92861019	0.494421191	-1.016176763	-1.684940447	0.5652328718
905520	2.92380175	-0.377000536	-0.886014238	-0.115243469	0.9848137147
905539	3.99267514	-0.959012260	0.204224433	-1.508902626	-1.7077458028
905557	-0.15500167	0.437623166	-0.461040030	-1.453254393	-0.8550878945
905680	0.93326963	2.104094210	1.431657320	-2.891717258	1.2742945854
905686	1.92792536	-0.891682104	0.042307522	-0.558392976	1.2792046734
905978	2.62003325	-2.499901968	2.432858840	-1.876362157	3.1480466996
90602302	-5.99833869	0.090949196	1.709697317	0.946542059	1.1740030698
906024	2.91930912	0.009204582	-0.900511128	1.789753754	-0.5764724361
906290	3.33900683	0.022205192	0.502858739	-1.141258134	1.5633857813
906539	2.01314368	-0.777767778	0.125468395	-0.659155792	0.0219104429
906564	-1.98671401	-2.314011967	1.235217081	1.768220225	-0.1438692164
906616	1.66813615	-0.861560770	-1.009417991	1.322731355	0.2813203167
906878	0.37085180	-0.113512779	-0.997485893	-0.155233295	-1.4746581730
907145	1.84733447	-2.538995601	2.003584668	0.483946194	3.5055609223
907367	4.69890766	0.431922385	0.005300424	-1.369403704	0.0850247711

907409	1.28205907	-2.548666752	-0.083786622	0.770833997	-0.5564209652
90745	2.31221232	-0.401678043	0.274381769	-1.668994639	0.5362392557
90769601	4.65498928	0.781611021	0.191953023	0.493247906	-0.5356508029
90769602	3.94000659	2.028231657	0.691127602	0.512954900	-0.3398155529
907914	-4.94550694	-3.003421435	-1.752529645	-0.671769430	-2.8622552978
907915	0.91499842	-2.476833776	0.362287911	0.261736218	-0.1383500423
908194	-4.54511193	0.815283142	-0.518299038	0.915534201	0.0714231738
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908469	2.19305464	1.803766114	-0.434726892	0.953104432	-0.5533135626
908489	-0.66385679	-0.436476802	-2.678425146	-0.728585540	0.0277629459
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909220	2.12331215	1.193746541	0.489940891	0.854644353	0.1058427332
909231	2.61301823	1.830449830	0.763657618	-0.822442197	0.3005228170
909410	3.20814515	2.233177743	-0.166078298	1.129064342	-1.2590809462
909411	0.30648073	-2.183077955	0.039251705	-0.486027975	-2.0876130896
909445	-2.47018342	1.498506224	0.590933256	-2.265358224	-1.4046042766
90944601	3.45168126	2.134583340	0.196989342	1.532828161	-0.9420463979
909777	3.89709781	-0.730133530	2.134193239	-0.780486142	0.8318195830
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9110732	-3.24256054	1.776795240	-1.791577273	-2.482874793	-0.5498682865
9110944	1.55585043	1.037014643	-1.300101127	0.621126067	0.3790198963
911150	1.24815985	1.587442008	-0.326126484	-0.924508658	-1.5340001134
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9111805	-2.50954497	2.526584455	0.797500590	-0.822950476	0.4288865709
9111843	2.21514321	-0.029865010	0.635900610	-3.373732618	-0.4795727546
911201	1.17279871	0.474422064	-0.481147553	2.142074589	0.2430866392
911202	2.83380799	1.017230750	-1.485323075	0.456004436	-0.1146041549
9112085	1.85545330	1.570011674	0.825756166	-3.926794310	0.2206325260
9112366	1.32744636	-0.776412499	1.255653387	-3.782215661	1.2439684480
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9113455	0.41729731	-0.116417262	-0.535671262	-0.363878888	-1.5121265752
9113514	3.83502421	-0.898388162	0.699948069	-0.219746414	1.3359873635
9113538	-5.92575082	-1.227216861	1.720074272	-0.473165926	-2.8810043579

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9113778	2.68148698	-1.442286469	0.339105123	-0.244601788	0.7919949893
9113816	2.03986918	0.902461858	2.840741857	-2.883294431	1.0158156443
911384	1.39806620	1.770668244	-1.478260847	1.306959009	-1.8037276563
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911391	1.98877953	-1.897792693	-2.029212586	0.911103793	-1.2304717877
911408	1.99642872	0.206189711	-1.639882150	1.038296903	-0.9034160493
911654	0.52020039	0.972512103	-0.323850426	-0.543521127	-1.4281800593
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912193	2.98286009	0.672747298	-0.589970772	-0.593827781	-0.4615631694
91227	2.47887396	2.361880555	-0.432234246	-0.383132558	-1.0372087673
912519	1.27099312	-0.509385643	-2.237023334	1.670906156	0.1175030515
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912600	0.05613371	0.227183495	-2.253532782	2.708217480	-0.8010805172
913063	-2.31535334	-4.385275755	1.776847691	0.968563355	-2.6430509823
913102	2.28232306	2.464672105	-0.820466562	-0.040767325	-1.0918206156
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913535	0.74112965	2.449879331	-1.995953713	-0.457537180	-0.2610613824
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91376702	1.65306732	4.551659142	0.751661685	1.646873743	-1.4018393993
914062	-3.10079133	1.235066483	1.286223339	0.608113830	1.1212732858
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914102	2.79087498	1.076940539	0.843799658	-1.109231547	-0.0048653882
914333	1.47362330	1.589695437	0.319710133	-0.798129881	-0.1579186103
914366	0.22919297	-1.514570755	-2.214334625	0.555081328	-0.2509810467
914580	2.55101378	0.763228924	-1.607988258	0.059875184	-0.5054003068
914769	-3.71243855	1.057755506	0.521494703	1.395773631	-0.3682903454
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915186	-1.29772469	-7.723057325	-0.688309017	2.890184346	2.0135831354
915276	-1.07574673	-8.287483337	0.433810751	1.952471847	1.3946448821
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915940	1.13058210	1.409761524	-0.564556376	1.993206904	-0.5280229629
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917896	0.61929383	-0.635789842	-0.488421490	-0.015141135	0.1063131129
917897	3.24945659	-1.284794677	-0.218284421	0.307537632	0.0993416275
91805	3.34997942	-2.670950443	-1.777523046	1.388364375	0.3541295063
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921362	1.14182718	-5.594535868	1.299893249	-2.186325923	-0.7649016389
921385	1.66401100	-2.387517361	1.500928782	0.875181074	-0.4841195986
921386	-1.01082308	-1.091429307	-0.632142163	-1.756972472	-1.0459061568
921644	1.29978604	1.819814057	0.372979316	-1.846544182	-0.1994515741
922296	2.37134219	1.680097929	0.384190094	-3.014076570	0.0382130321
922297	1.66440651	0.213774641	-0.147942249	-0.196878516	-0.5559910407
922576	1.92598353	1.136739705	0.477781900	-1.156482818	0.2021640255
922577	4.23349159	-0.184110499	-0.326131445	0.587786142	0.7040073381
922840	2.67551655	-2.313756961	-0.053800867	0.340150357	-0.4502519824
923169	3.83312511	-0.495813665	0.922428047	-0.551386509	-0.0799652223
923465	2.54919727	-0.228129228	1.412934778	-1.969057256	1.4550448383
923748	4.69079604	0.766803238	1.542607509	-0.778334167	0.7842069600
923780	2.02325691	-1.260133116	0.504482203	-1.134529125	0.1141621816
924084	2.89340232	1.450359601	0.779859803	-2.967836815	0.3568359788
924342	3.49912218	-1.799249342	2.764024589	-0.865545694	0.6248149194
924632	2.15201013	0.829339088	0.564300199	-3.009108230	-0.5653278607

924934	2.05327740	-1.615038205	1.837342797	-3.110797801	-0.2108571316
924964	3.87388097	-1.083301553	1.858308794	-0.433358360	1.6859553304
925236	4.06028949	-0.122061034	3.235925374	-3.466133119	2.2399371646
925277	0.09858059	0.213372093	0.388587548	-1.011820230	-2.9864399976
925291	1.08841850	-1.291711328	1.428122289	-3.369171941	-0.9465087393
925292	0.48134743	0.177863190	1.031200235	-2.008512908	-0.3268156112
925311	4.86602793	2.129232607	3.411187296	-5.129474909	0.4884909563
925622	-5.91241029	-3.479575000	-3.259923297	-3.914141413	-1.7148732547
926125	-8.73365338	0.573350185	0.896301447	0.384811315	-0.6784463175
926424	-6.43365455	3.573672989	2.457324373	1.176279049	0.0747585153
926682	-3.79004753	3.580897052	2.086640366	-2.503824654	0.5102735378
926954	-1.25507494	1.900624364	0.562235817	-2.087390339	-1.8084001296
927241	-10.36567336	-1.670540206	-1.875379194	-2.353959892	0.0337122683
92751	5.47042990	0.670047220	1.489132801	-2.297135901	0.1845409324

	PC6	PC7	PC8	PC9	PC10
842302	1.410183639	2.1574715203	0.398056982	-0.156980233	-0.8766305374
842517	0.028631162	0.0133463502	-0.240776600	-0.711278966	1.1060217613
84300903	0.540976145	-0.6675790845	-0.097288135	0.024044486	0.4538760367
84348301	3.050737497	1.4286536346	-1.058633755	-1.404204115	-1.1159933184
84358402	-1.225416405	-0.9353895017	-0.635816609	-0.263573547	0.3773724202
843786	-0.450642134	0.4900139551	0.165298432	-0.133355760	-0.5299649041
844359	-0.128835073	-0.3013023315	-0.083624609	-0.079954220	0.2189506984
84458202	-1.255934103	0.9732434974	0.652763899	0.247965464	0.9997064885
844981	0.559052820	-0.2149145974	0.686736938	0.511473855	0.0291618051
84501001	-0.723272337	2.5176245802	-0.364827879	-0.716766009	-1.1646060593
845636	0.127219899	0.6244628561	-0.165603833	-0.301115608	0.3759046163
84610002	0.764194227	0.6655068738	-0.619000411	-0.400588340	-0.3307781202
846226	2.592290301	0.9406696600	3.318863750	-0.747130974	0.3955980681
846381	1.006774257	-1.7974995418	-1.105684467	0.267389882	0.4391951701
84667401	-0.349408795	0.7422274132	0.280679206	-0.674510305	-0.1155553104
84799002	0.495342130	0.9142524133	-0.302257713	0.456538430	0.4467643551
848406	-0.761520960	-0.0867706702	-0.029483040	0.502783775	-0.4992331367
84862001	-0.751827782	0.4415962916	0.413405991	0.018656683	0.3563165037
849014	-1.136688687	-0.2510619539	-0.741499503	0.410086322	-0.5708248435
8510426	-0.048594023	-1.2006310194	-0.403270875	0.404944121	0.0059203674
8510653	0.430462491	0.3381860916	0.861201670	-0.187036602	0.3606065559
8510824	-0.769391365	-0.1908707908	0.244247147	0.546415425	0.0076299444
8511133	3.020386239	-1.0062943167	-0.104544724	0.168363304	-0.2454047607
851509	-0.595316618	-0.0747626295	-0.180094245	-0.363806008	0.6909368005
852552	-0.812165001	0.7996017629	-0.419539913	0.810965561	0.0214595144
852631	1.183160422	0.6013244034	1.306262282	1.100042783	2.2842657360
852763	0.786665153	0.2896641020	0.495541562	0.113090873	0.2951812417
852781	-0.906797174	-0.6452669891	-0.340174492	-0.043873679	-0.3961957456

852973	0.192679233	0.1609885641	-0.378501497	0.356974453	-0.5691360315
853201	0.036227947	0.1838685926	0.218709799	-0.381333094	-0.0364970759
853401	0.241867529	0.4615706215	0.291813749	0.284492214	0.1182178679
853612	1.233593969	1.9763494638	-0.202453666	0.564148659	-0.2297663753
85382601	-0.190839648	-0.8888020606	0.243842164	0.442519010	0.1433737325
854002	0.267243832	1.0317090641	-1.141727259	-0.566293378	0.2446991603
854039	1.000905829	0.7700944229	-0.210139235	-0.585132155	-0.0747302216
854253	2.009267717	-0.7199616429	-1.411392236	0.196789071	-0.7431027433
854268	0.568095150	-0.1425954318	-0.488198042	0.390894097	-0.3824484486
854941	0.290013739	-0.8043318792	1.707341340	-0.891664549	-1.0876247249
855133	-1.493142577	1.2791292757	1.498249655	2.100532774	0.0114924931
855138	-1.036438899	-0.1775681622	-0.198083551	0.431153558	-0.4385877226
855167	0.928295680	0.0418357439	0.048599880	0.666574547	0.3902914311
855563	-2.120772389	-0.4965417472	0.484027443	0.912512891	-0.4303234931
855625	4.307303747	0.4337837927	-1.387017258	-0.352517752	-0.9339783433
856106	0.242568260	0.6132339088	0.067809257	0.282512565	0.1518262957
85638502	0.110049368	0.2333988676	-0.174872997	0.544441256	-0.2067359385
857010	0.423532149	-0.7874816331	-0.790091300	0.111457042	-1.0551322678
85713702	0.575420451	0.2230057137	-0.867569780	0.328258562	-0.1456716219
85715	-0.281861620	-0.1604792421	-0.018943396	0.591809797	-0.1053337966
857155	-0.692882067	0.1971605392	0.040044188	0.419441177	-0.6168907916
857156	0.544804546	-0.6838772921	0.555862941	0.605703404	0.1987895162
857343	0.266013819	0.4536379429	-0.123851495	0.259381187	-0.4343429520
857373	-0.256313209	0.4449438979	0.057879905	-0.492076521	-0.2410267010
857374	0.629337135	0.0607736151	-0.510873270	0.413623505	0.8230489282
857392	0.707227400	0.6479398411	1.537761700	-0.614290601	0.0886086788
857438	-0.656077776	-0.2152637743	-0.207515896	0.505338373	0.1085018162
85759902	1.277739611	-0.3925629141	-0.745416793	0.301153902	0.0751610030
857637	-0.178434830	-0.1853434904	-0.309233059	0.012534355	-0.1128753503
857793	0.461927636	0.5362098378	0.951178388	0.073379908	0.1941817154
857810	0.705119501	0.4873781448	-0.297723909	-0.127261763	0.7362638224
858477	-0.900267281	0.4064726785	-1.225062124	0.345609537	-0.4642608787
858970	2.538550035	0.7457813449	0.909941133	0.262199059	1.3178790775
858981	-0.815018482	-0.3959097935	-0.230734976	0.102956869	-0.9272860914
858986	-1.218994501	0.8193030850	-0.330742314	0.528189577	-0.4855301241
859196	2.955682829	0.2103798445	0.627427082	-0.232120660	0.0413657355
85922302	-1.242117920	-0.0057834579	-0.385706296	0.801372546	-0.1817471124
859283	-0.727422417	-0.5309593004	0.474393258	0.003939003	0.1728882560
859464	-1.188527693	0.1693780934	-0.029824460	-0.111145907	-0.2713131090
859465	-0.405353080	-0.5265287503	-1.082071976	0.427426065	-0.3912329742
859471	2.213816387	-2.5787909992	-1.298010892	3.780791875	-2.7357511959
859487	-0.433947228	-0.8493285940	-0.334869937	-0.004978691	-0.3104418547
859575	-0.291543973	0.2832433298	-0.086188560	0.326867199	0.0333168740

859711	-1.638316786	2.4262266414	-1.317108728	-1.883468421	3.1898325590
859717	-0.539586813	1.5624901202	-0.593662500	-0.672507942	0.0611282077
859983	-0.919638446	0.6971724884	0.393036492	-0.357905538	-0.1474184494
8610175	0.034838302	0.1200189004	0.459971988	0.314694924	-0.0155400843
8610404	-0.994688029	-0.9081210535	-1.100360740	1.244777578	0.0926553152
8610629	-0.612562561	-0.7756714142	0.246838044	-0.772831741	1.5166900630
8610637	0.682870529	1.3985754634	-0.719241167	-0.096664805	0.5085691762
8610862	4.876508199	-1.1544779461	-0.483077495	-1.170391748	-1.3237161139
8610908	-0.072350138	0.0536554932	0.355866547	0.048832592	-0.1077491295
861103	-1.501869476	0.1435614258	0.486258862	-0.122378079	-0.0390292436
8611161	0.138012244	-0.3263920580	0.151919734	-0.226864771	-0.5036154424
8611555	-2.186236690	0.0355707110	0.547079286	-1.647859246	0.3961746350
8611792	-1.421297891	-0.0432176692	2.012767522	-2.966728252	0.0222412116
8612080	0.882263930	-0.2550878307	0.229994161	0.423649141	0.1477716874
8612399	0.967176566	-0.1536701262	0.192125163	-0.083450911	0.4426520533
86135501	0.567360307	-1.6613223577	0.349250281	0.571523884	-0.1210561314
86135502	1.560091353	-0.1983680707	-0.643025712	0.183841722	0.4623711739
861597	0.223702425	0.0769126070	0.148776905	0.262252261	0.4555666494
861598	0.842299884	-0.1736551116	0.614011283	0.283254129	0.3987023515
861648	0.325518164	0.2995505173	0.128752056	-0.246304891	0.5210683990
861799	-0.199459121	-1.7255946363	-0.786058401	0.116745956	0.1441268207
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862009	-0.321113861	-0.5677543179	0.395718555	0.058107568	-0.4912485698
862028	-1.066707998	-1.0421552988	-0.283564277	0.637899776	-0.2969722678
86208	1.783555798	0.0004244298	-0.292506839	-0.580370555	0.4991648297
86211	-0.096852967	-0.0225249946	1.596299697	0.179428829	0.4099629648
862261	-2.351934315	1.1157841763	0.148452301	-0.906614963	-0.3165086702
862485	-0.430903499	0.4844971328	-0.290694588	0.057293175	-0.3825068496
862548	-0.853188272	-0.5060704070	0.634423963	-0.237141676	0.1392901935
862717	-0.595802895	0.0029648840	0.062601958	0.797723981	-0.2095356865
862722	-0.899700995	1.4116521142	0.324334412	-0.505544952	-0.1898542596
862965	0.342937555	-0.2377974400	0.513345470	0.454774516	0.1528759939
862980	-0.991047162	-0.5084776263	-0.051816347	0.885904532	0.1673165990
862989	0.940750946	0.1577998775	0.592437419	0.477922672	0.6483605373
863030	-2.202856495	0.5190182681	0.919491499	0.024774004	-0.5966611023
863031	-1.608771501	-0.4979118440	0.452844869	0.527102050	-0.4592233167
863270	0.359599477	0.1231328050	0.263560697	0.289616641	-0.1759008102
86355	1.020860934	-0.0954075198	0.767942841	0.627017416	-1.0979068211
864018	-1.439608866	0.2115695617	-1.388075349	0.448512027	-0.6290141193
864033	-1.095444306	0.2037178296	-1.218964989	-0.570481422	-0.7426654271
86408	-1.458702835	-1.0401187280	0.120431563	-0.227163027	0.3361622567
86409	0.833123888	0.4111745398	0.970303417	-0.535303964	-0.2484248020
864292	-1.261777184	0.1057625720	0.713697894	-0.410313572	0.6957360576

864496	-1.706440486	0.5119879213	-0.337466388	0.678858808	-0.4910440978
864685	-1.337752701	0.4595072608	-0.488509104	0.354100824	0.4414446401
864726	-2.293687595	-0.9185965070	-1.898086528	0.154241095	-0.4446864733
864729	-0.767555623	-0.2067958374	0.209629326	0.284564474	-0.0334444954
864877	-1.492492721	0.4264319465	-0.275527717	0.357573419	0.0964420085
865128	4.975158195	-0.9917687699	-1.044205851	-0.483438131	-0.3707162449
865137	-0.438000429	0.0569341545	-0.463647307	0.245730642	-0.4929124233
86517	-0.579613971	-0.2505498519	0.361729650	-0.241886354	0.1402872611
865423	-0.946220851	-1.2957391097	-0.792061520	-1.694971898	-0.0776531867
865432	0.180693692	-0.5030415897	1.294717404	-0.091526472	-0.3633815227
865468	-0.247250211	-0.1912870373	-0.070698364	-0.197988493	-0.4727441430
86561	-0.226920405	0.3213445519	0.269963020	-0.171115561	0.2946382608
866083	0.148751350	-0.0415938395	-0.272031126	0.771906177	-0.0588503290
866203	1.467236870	-0.1445960967	0.172105084	0.253497334	0.2448693537
866458	-0.533827466	-1.2652594966	0.721508750	-0.385475878	0.5934329421
866674	0.678064911	-0.7403080258	0.634019823	-0.195963883	0.7616033847
866714	0.654051923	0.3144589921	-0.222639255	-0.683091883	-0.0068671508
8670	-0.785225256	-0.6210090246	0.214754724	0.669006569	0.1309451308
86730502	0.686370956	-0.3132321125	0.524909363	0.333859949	0.7119644915
867387	0.246141926	-0.8388130034	0.525219945	-0.232429603	0.2414018451
867739	-0.384108551	-0.1387712566	-0.400629913	0.015768321	-0.1420031613
868202	-0.711654240	-0.3170822284	-0.423054123	0.296085713	-0.1660916070
868223	-2.039215085	-0.0047169897	1.745367912	0.054013249	-1.3406194472
868682	0.067229553	-0.4386830029	0.624428298	0.431789796	-0.1857668917
868826	1.472940622	-0.5628536813	-0.555419588	1.433116172	-1.8906385275
868871	-1.755538141	-1.3946449979	0.617942137	0.193853924	0.0783272296
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869104	-0.173947054	0.5618521558	0.297047282	0.162636806	0.0261558272
869218	-0.778202592	0.1691115541	0.976926700	-0.507861462	-0.6639930913
869224	1.073297753	0.0989681910	-0.372823313	-0.071154664	-0.2542790364
869254	-0.099111239	0.3467449413	-0.147442289	-0.015463023	-0.7720259012
869476	-0.382201396	1.3607862474	-0.371332854	-1.206190790	1.1885364740
869691	5.129714191	-0.7872305021	0.097521762	-0.086192705	-0.6385226560
86973701	1.553852814	0.6731568921	0.319662491	-1.417845937	0.0564845959
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87106	0.154082267	-0.5847800143	-0.491749917	0.213116694	-0.0635198173
8711002	0.970037522	-0.4276196991	-1.276188474	-0.121835356	-0.4266320967
8711003	0.864745584	0.1069268147	0.109240323	0.453530595	0.4909214506
8711202	-1.035185529	-0.3174416715	0.420352680	0.213503999	0.2869782720

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871149	0.153281267	0.9054318895	-0.623447958	-0.044924496	-0.3461897877
8711561	0.588778074	-0.0348880820	-0.015419238	0.186223933	-0.2572771820
8711803	0.588871013	-0.8310914625	-1.017671016	0.523162928	-0.0911605179
871201	0.343422459	-0.0707133402	0.641697510	-0.013645624	-0.3983688340
8712064	-1.889969783	0.0290416393	-0.417881679	-0.629496118	0.0577694399
8712289	1.042121842	-0.0704226922	-0.832185834	-1.088425879	0.7422035975
8712291	0.723455982	-0.4223737010	0.138398073	-0.455293689	-0.0506963170
87127	-1.686598428	0.2821154593	-0.372316236	-0.110448647	-0.6643564473
8712729	0.626105499	-0.3516309352	0.257336906	0.098062925	0.3097561544
8712766	-1.976653127	1.5974667481	0.479005437	0.658932179	-0.5988452380
8712853	-0.283614915	-0.5536495883	0.912792548	-0.223496888	0.3784836258
87139402	-0.044220048	-1.0475041547	0.026274210	0.505731618	0.3347146313
87163	-0.422267087	0.0357420256	-0.352737039	0.808130669	-0.4802621497
87164	-0.565776004	-0.2672482462	1.064477530	-0.093245598	-0.6385242481
871641	-1.342695187	-1.0265266652	-0.442934433	-1.288967588	-0.2346218069
871642	1.346133180	0.2723286751	0.164406007	-0.669318625	-0.5370750064
872113	0.220842652	1.1241630693	-0.637406602	-0.424652143	0.0577518712
872608	-0.388805486	0.7135687514	-0.502118964	-1.248884797	0.1510833104
87281702	-0.268706635	-0.9944966018	0.217537500	-0.231151234	-0.4710279564
873357	0.925966830	0.4519886400	-0.091733964	-0.179338922	0.1021444906
873586	-1.507807056	0.3330008518	0.077576227	-0.396246129	0.2634730538
873592	-0.835878816	-0.7140340991	0.801923303	-2.159841626	0.0290996051
873593	0.725136194	0.8451468587	0.591019652	-1.014728770	1.2015974832
873701	0.094248735	-0.0607585087	-0.365255024	-0.120588274	-0.4112053722
873843	-1.003522807	-0.2666554566	-0.089381943	-0.084952464	-1.1914664073
873885	0.183576123	0.6909743372	-0.214640324	-0.096784999	0.4605384186
874158	-0.793026617	0.4370590422	-1.900279550	-0.315356103	-0.1041326653
874217	0.509781798	-0.5622981929	-0.235231929	-0.325336092	0.1245710431
874373	-0.634963902	-0.5973256464	-0.160681786	0.540712933	-0.5131869412
874662	0.695238561	-0.9098868735	0.019838578	-0.533804197	-0.9625494330
874839	0.692017491	-0.1875842559	-0.316611583	0.063671760	-0.2430371800
874858	3.668891109	0.4667888874	-1.321343281	-1.944665503	-0.9460823770
875093	1.004175357	0.9998306814	0.995368772	0.232326073	-0.1188888078
875099	2.091661791	1.1663610596	3.949687763	-2.059344309	-2.6696617876
875263	-1.200309594	0.6282875359	-0.457500580	0.252184555	0.0222141330
87556202	0.005813478	-0.8631588057	-0.204826688	-0.356562840	-0.4112333371
875878	1.207398464	-0.3659380358	-0.113716688	0.329933369	0.1978627710
875938	-1.698608135	-0.9030444488	-0.446979767	0.339740677	-0.7976099029
877159	1.216994994	-0.0181619840	-0.178446492	-0.124884357	0.5853707934
877486	0.321490348	0.1679841528	-0.668695410	-0.891981952	0.3036968071
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877501	-0.540711745	-0.1667847179	0.291643165	0.597355421	0.6557436271
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878796	-0.331940888	-2.6682527030	0.260934777	-0.462552726	0.4348839944
87880	-1.178677385	0.0130692386	0.157838548	0.532754157	-0.2797121016
87930	-0.160348396	0.1870705062	0.068129717	0.533288806	0.1893951033
879523	0.042891323	0.3408804873	-1.098677759	-0.280446322	-0.2526169016
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8810158	1.188797426	0.8614230375	-0.268868458	-0.719578899	0.0280288685
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881046502	0.984117981	-1.5402069105	-0.537234103	0.409349077	-0.6771736584
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881094802	-4.214059597	-4.0810933903	-4.358470616	-0.658094683	1.7059309508
8810955	2.462907510	-0.7760274006	-1.521890501	-0.521797791	-0.2906150561
8810987	0.568593508	-0.0375813211	0.444521903	-0.033533870	0.0725430982
8811523	0.849146479	-0.9034469827	-0.408260911	0.385297416	-0.2190552567
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9011494	0.988060282	-0.7970441849	-0.245890513	-0.272785401	-0.3651742847
9011495	0.398140618	-0.0645026624	-0.560973305	0.467678796	-0.3926243393
9011971	-0.832344744	1.4919773252	-0.496542165	0.708345052	-0.0169532931
9012000	-0.477923128	0.8442636938	0.636164367	-0.204859009	0.1625337312
9012315	2.667823055	-0.9407899084	-0.869846292	0.423379905	-0.4379865500
9012568	0.617275952	-0.4793053363	-0.220678981	-0.323424153	0.3717467237

9012795	0.253631779	-1.1782462018	1.518653064	-1.823630553	0.5338030161
901288	-1.030121857	-0.7467180332	-0.695323516	-0.378361717	-0.3918087362
9013005	1.617443807	0.0938490707	-0.344280981	-0.066635392	0.5216969893
901303	0.775977838	-0.2614574286	0.536962646	-0.783668674	1.0015978141
901315	-0.062587840	-0.3217196360	0.885498583	1.237510046	1.2853970174
9013579	0.048936764	0.3139548052	-0.557032840	0.002152141	0.3847686640
9013594	1.034252471	0.2979977108	-0.774880424	-0.414667982	0.3611762162
9013838	-0.931622121	0.5558930985	-0.315289940	0.568236652	-1.3305107926
901549	-0.413238965	-0.3265681118	0.841546961	0.281467270	-0.2092956126
901836	1.511904788	0.1107862663	0.294876597	0.346448987	0.2913150267
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90251	-0.535690225	0.8239055160	-0.169622036	-0.256319858	0.1735146645
902727	0.514203265	-0.1400589671	-0.134319768	0.133553755	-0.3932263761
90291	-0.634221424	-1.2036581988	-0.193479374	0.611419040	-0.2077215786
902975	0.836900468	0.5372471084	0.120920306	0.128158478	-0.3976112633
902976	1.269631553	0.3369059661	-0.144015062	0.021957292	0.2938631449
903011	-0.903282021	0.7048671644	-0.198298707	0.173642809	1.4348701722
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90317302	0.156093205	0.3242076866	0.190087018	0.241888734	-0.0729502673
903483	-1.848848148	1.5475878305	-0.444033150	-0.620675476	0.8826008510
903507	-1.016195655	-0.0050197039	0.316577899	0.564083320	-0.6156462037
903516	0.485646865	-0.4029431540	0.372494053	-0.926441986	0.1201789382
903554	-0.076639013	-0.5411420517	0.397793364	0.113082074	-0.1676621390
903811	0.386466855	-0.6426712317	0.749554481	-0.569956095	-0.1439670700
90401601	-0.744129779	-1.0668559939	0.971925397	0.503561293	-0.2890358887
90401602	0.203701388	-0.0995017006	0.539132750	0.148387464	-0.2783778826
904302	0.073042200	1.2376399689	-0.109921920	0.120093235	-0.1420028943
904357	0.168954300	0.6871762127	0.258398388	0.479129168	0.2528963004
90439701	-1.256879923	-0.2859010637	0.218312252	-0.496045403	-0.4166135405
904647	-0.691195823	-0.0520373351	0.085710694	0.268108418	-0.5865594042
904689	2.427978134	0.2056036891	0.168971978	-0.140625905	0.1210694614
9047	0.937091754	0.0225694328	0.698185070	-0.069564406	-0.2315344151
904969	0.596538718	-0.2936325230	-0.407229995	-0.043375151	-0.6090000775
904971	-1.417278885	0.0553933955	0.185171279	-0.011947767	-0.9140086363
905189	0.304102683	-0.5350338384	0.436699305	-0.283793452	0.1341185165
905190	0.930130928	0.7878753397	0.029225670	-0.455179924	0.0062052408
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905557	1.241782183	0.1692927893	0.303507283	-0.023140989	0.4959808194
905680	1.625262058	-0.5471226180	-0.849327394	0.368361131	0.2919367892

905686	0.116726716	-0.4680111924	-0.667070503	0.265822576	0.6191312844
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924632	-0.549918834	-0.4839795902	-0.892397382	0.435103530	0.2439245685
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924964	-0.814176294	-0.4706597676	-0.071321985	-0.701263764	-0.0705817127
925236	0.174755382	1.0763071056	-0.129380621	-0.426655069	-0.2770426795
925277	0.012441261	-0.2900018002	0.312069081	-0.067804247	-0.3254840515
925291	-1.724352624	-0.3536636324	1.163315907	-0.207312716	-1.3904847631
925292	-0.991190143	-0.0485030055	0.195049336	-0.325578857	0.4286949972
925311	-1.109815814	0.4879418774	1.288150901	-1.389256180	-1.7451271378
925622	1.012707855	0.2712037562	-0.091882489	0.372148526	-0.6298691742
926125	-0.300488375	0.0947410215	0.733436507	0.830362739	0.4502424273
926424	-2.373104852	-0.5956056075	0.035439747	0.987060511	0.2567626672
926682	-0.246493221	-0.7156966020	1.112380784	-0.105114917	-0.1085369986
926954	-0.533977345	-0.1925887705	-0.341586475	0.393570521	0.5204194692
927241	0.567437190	0.2228855560	0.279992246	-0.541558028	-0.0892176149
92751	1.616415088	1.6974579729	-1.045433644	0.373772231	-0.0476836145

	PC11	PC12	PC13	PC14	PC15
842302	-0.262724285	-0.8582592992	0.103296775	-0.6901967966	0.601264078
842517	-0.812404827	0.1577837523	-0.942699807	-0.6529008436	-0.008966977
84300903	0.605071514	0.1242777100	-0.410265611	0.0166650952	-0.482994760
84348301	1.150501182	1.0104267377	-0.932450704	-0.4869883986	0.168699395
84358402	-0.650787016	-0.1104182769	0.387606915	-0.5387065429	-0.310046684
843786	-0.109669824	0.0813699015	-0.026251351	0.0031339435	-0.178447576
844359	-0.393343607	0.2419761651	-0.351941936	-0.2216015267	0.060855369
84458202	0.196313828	-0.2777917914	-0.460630835	-0.0961356710	0.187478261
844981	0.560021577	0.0788169039	0.493001875	0.1120530076	-0.034237132
84501001	-0.548555069	-1.0841629631	-0.375475245	-0.0006033716	-1.206141020
845636	-0.294276512	0.6099198621	-0.692298563	-0.1301140961	0.091099184
84610002	0.150099383	-0.8725334105	-0.978852628	0.4132851288	-0.076575493
846226	1.119190867	0.0520074379	-1.983208918	-0.2667913889	0.492408131
846381	0.853628961	0.0628135272	0.194455743	-0.1212264306	-0.151174299
84667401	0.637371253	-0.5439530565	0.152152510	0.0208341513	-0.504094726
84799002	-0.089892055	-0.0929155801	0.272937688	0.4193736451	-0.531937246
848406	-0.283115330	0.3228336349	-0.873313571	-0.2068809463	0.026977460

84862001	-0.077022788	0.1450773283	-0.056826726	-0.5557727446	0.357788221
849014	-0.382631859	0.0680073259	-0.762829689	-0.1781151366	0.186075363
8510426	-0.087947737	-0.3232036086	-0.240304220	-0.1497219434	-0.277740235
8510653	0.252743308	0.1820993618	0.534747411	0.5364927035	0.630306593
8510824	0.295829357	-0.1193713638	-0.105109885	-0.5505241800	0.208888166
8511133	0.747761739	-0.7897394650	0.018967583	-0.3939657391	0.178368816
851509	-0.845138812	0.1995874763	-0.760017755	0.1495561475	0.585041105
852552	-0.600095326	0.2321645154	-0.719018102	0.0560160318	0.622570169
852631	0.196585356	-0.8095860291	-0.394727262	-0.9305675819	-0.282237083
852763	0.441362589	-0.9739851395	-0.516422067	-0.7074323955	-0.383724915
852781	-0.412668030	-0.3158058348	-0.298472337	-0.2950812415	-0.235804579
852973	0.370382886	-0.1246372082	0.129900549	0.1958202715	0.175956263
853201	0.246904961	-0.0031839923	-0.222279620	0.1283529520	0.091747519
853401	-0.496378073	-0.1044630774	0.885636344	0.0692396351	-0.376293016
853612	-0.756864461	-0.2657615457	-0.522186276	-0.3000453967	0.072356014
85382601	-0.585836567	0.2547032178	1.119529771	0.2184706425	-0.403789163
854002	0.077919478	-0.1992975316	0.307173136	0.2036374012	0.068495324
854039	-0.326307877	-0.4066154950	-0.351748055	-0.0564880712	-0.224784255
854253	-0.328257659	0.2292843967	-0.619894462	0.2143138994	-0.067766013
854268	-0.217741168	-0.0195015091	0.032551005	0.1704452228	-0.479641248
854941	-0.032328754	0.4115801153	-0.245663247	-0.3627028769	-0.150524126
855133	1.047288688	-0.6130946673	-0.290379074	0.1249868104	-0.401912364
855138	0.591014507	-0.0004681369	-0.235544182	-0.4578511130	-0.668135575
855167	0.145009410	-0.0423201249	0.029903294	0.0054147710	0.005022251
855563	-0.115306889	0.3312318699	0.019742154	0.2321540822	-0.128765770
855625	1.277954488	-0.8309410010	-0.922742372	0.9892697016	0.220725927
856106	0.160272727	0.1311301054	-0.144226466	-0.1816995714	0.160135565
85638502	0.240085901	0.1091591008	-0.164573360	-0.0950096841	-0.193897144
857010	0.003089503	-0.1206287088	-0.149769253	0.3143049855	-0.268535008
85713702	0.090090936	-0.0249037139	0.378695789	-0.2394451655	0.277696370
85715	-0.289430944	-0.0732252281	-0.561084977	-0.2625113637	-0.646184357
857155	-0.324354020	0.0931121402	0.393569102	0.5995102295	-0.169047859
857156	0.237322628	-0.0116411885	-0.208908133	-0.1508619753	-0.058578846
857343	0.639460669	0.6058597387	0.224644982	-0.0446479438	-0.040260858
857373	0.079686662	0.0240053741	-0.377273811	-0.4453178247	0.108861412
857374	0.025165467	0.1593213356	0.246020193	-0.5093448931	-0.077987883
857392	-0.274018692	0.6533266173	0.601165133	0.6072878098	-0.501815152
857438	0.058487255	0.2677384961	-0.399395210	-0.0450747265	-0.036176457
85759902	0.197754405	0.5401834853	0.058602197	-0.1399689340	-0.349863814
857637	-0.605550553	-0.1156566994	-1.096994917	0.1275301954	0.597982425
857793	0.381160012	0.2814152874	-0.376294366	-0.0229779469	-0.331463457
857810	-0.228334046	-0.1991733623	0.421272292	0.2167665417	-0.342304785
858477	-0.621205924	0.3367435600	-0.148838932	-0.4425445128	0.511520357

858970	-0.369405174	0.4379929489	0.090549052	0.3353742675	-0.536697760
858981	0.341145747	0.3759209730	0.161076437	1.1878815701	0.321697375
858986	0.717323304	-1.1859167706	0.589385595	0.3030289831	0.129915286
859196	-0.511053316	-1.0091261525	-0.229682636	-0.7934928857	0.139211976
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859283	0.510796559	0.5421158683	0.129143047	0.1548683357	-0.008334256
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859471	-2.301916714	1.9478812125	1.103679434	0.2450717685	1.060326663
859487	0.131584973	0.5400548609	0.277961017	0.1141297357	-0.357009919
859575	0.428848750	0.2049555851	-0.371823339	-0.1793614799	0.291771882
859711	-0.057789430	0.5105820620	-0.335217803	0.4799300038	-0.246973803
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859983	0.379838236	-0.1163449791	-0.053581681	-0.1557475752	-0.059940319
8610175	-0.007559389	-0.2732435321	0.012174976	0.2578796832	-0.032393640
8610404	0.140571180	-0.7691753712	-0.428252247	-0.1639715182	-0.078397551
8610629	-0.408291577	-0.3217815151	0.814890878	0.8570418853	-0.590679517
8610637	0.845106205	-0.7038553721	-0.692840046	0.3237751499	-0.106235419
8610862	1.209082105	1.2860572446	1.339239602	-0.5384927467	0.904755029
8610908	0.127065020	0.1355485485	0.028337746	0.5169396446	-0.020759654
861103	-0.786967580	0.0345574874	0.033439840	0.1843125970	0.447952176
8611161	0.195944932	-0.3957884419	-0.141772499	-0.0148270384	-0.153420304
8611555	0.253381955	-0.5690714729	0.753989687	-0.5460281371	0.432021867
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86135501	-0.552045457	-0.8029870413	-0.067392082	0.1127709163	-0.092173983
86135502	-0.433326769	0.5703689588	-0.119188327	-0.4170696748	-0.396379852
861597	0.016642280	-0.3158031850	-0.382598138	-0.4374455309	-0.208840242
861598	0.741940542	0.0200969537	-0.065901136	0.8226864857	-0.236364320
861648	0.724144967	0.5554322605	0.214432986	0.3277344219	-0.037958812
861799	0.937374393	0.4067146390	0.563287417	-0.7725393225	0.029255733
861853	0.049407605	-0.6890916537	-0.464225889	-0.4063362427	0.014895270
862009	0.148850187	0.1462808064	-0.128365565	0.7090543979	0.102181427
862028	0.609344020	-0.7423625667	0.222233620	-0.2428698410	-0.129345613
86208	-0.644167720	-0.1613821271	-0.079839276	0.2971928632	-0.073910222
86211	0.585191831	0.4438091073	-0.230398088	-0.1827187582	-0.061170624
862261	-0.329894644	0.3323522602	-0.138254531	-0.1072538655	-0.178613587
862485	-0.146550183	0.0692879791	-0.002258044	-0.4559046799	0.100650887
862548	-0.407700886	-0.7112958461	-0.473288604	-0.2122558723	-0.075410969
862717	0.237747713	0.4271403452	0.204387799	-0.0197655248	-0.095573905
862722	-0.610239332	0.4132517311	0.433086920	0.1177467385	0.467007135
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863030	-0.564911664	0.9193590685	0.012370125	-0.2168197204	-0.865838150
863031	-0.154806897	-0.1866572122	-0.633643189	0.4039318163	-0.108589053
863270	0.066405975	0.0760311613	0.128667473	-0.0370781851	0.297765381
86355	0.090596486	-0.7421530470	1.075751502	0.9368104783	0.533299888
864018	-0.320799033	0.1800812978	0.032265835	-0.3023335385	0.215077193
864033	0.327746967	0.2046597881	0.120237817	-0.1783615485	0.194251397
86408	0.650777643	-0.2933259538	-0.112544389	-0.7586715299	0.283199791
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864292	0.313444155	0.1805138306	0.390237981	-0.1019964618	0.592050290
864496	0.287039611	0.8103129309	-0.230171199	-0.1255945235	0.341881787
864685	0.262233097	-0.0114082847	0.074866229	0.6410038747	-0.136567267
864726	1.325817930	-0.5164890912	0.796713304	-0.1042779720	1.230334163
864729	-0.350864753	-0.3624834189	-0.267331949	-0.0465436932	-0.121090198
864877	-0.621096064	0.0271540484	0.412231332	-0.2522319704	-0.251715646
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86517	-0.884232338	-0.0265148144	-0.611872230	-0.2217926975	-0.115027379
865423	-1.605406626	-1.1915246645	2.585042825	1.4099705217	-0.170480479
865432	-0.169709366	0.4297105020	-0.047285037	0.0122072432	-0.036636413
865468	-0.362363688	-0.6734598556	0.401537066	-0.1036802249	-0.014825254
86561	-0.110403142	0.1048734656	0.245746438	0.2803976055	0.304557047
866083	-0.072071812	0.3284031569	-0.153101749	0.1037941218	0.303579465
866203	-0.527253976	-0.5854540578	-0.196247938	0.4671088272	-0.058378954
866458	1.402586521	-0.2463677463	-0.066270509	0.2971831513	-0.052689797
866674	-0.572576039	0.0549812995	1.581189240	-0.0155871349	-0.138038169
866714	0.126816678	0.1476258572	-0.512886476	0.5304867215	-0.094368210
8670	-0.107029230	0.2008584867	0.446006724	0.1601414909	-0.417882075
86730502	-0.317576548	-0.3185102735	0.169930917	0.5385371207	0.257834920
867387	0.108520078	-0.2176028380	-0.246663060	-0.1122800805	-0.130306412
867739	-0.789606706	0.3201095342	-0.079950656	0.0860593677	0.240075552
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868223	-0.353033591	-0.1571192120	-0.825311501	0.3718701966	-0.226420414
868682	0.236340337	-0.2435946108	0.154797172	0.2682689068	-0.072006787
868826	1.531930455	0.7707769845	-0.892271606	-0.2731473901	-0.350272945
868871	0.970602631	-0.9099026342	-0.218520602	0.6656889737	-0.102185136
868999	-0.575442560	0.3454684410	0.439023367	0.1286825140	-0.141030598
869104	-0.343156256	-0.1147671451	-0.111714903	-0.1119664738	0.188766533
869218	0.089613812	0.2591418464	-0.512083663	0.5144532498	0.227154614
869224	0.036775131	-0.1562905573	-0.128441182	-0.2541570849	0.301604642
869254	-0.172759701	-0.4754195170	0.207302817	0.0873913224	0.051752943
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871149	-0.398844589	-0.1954767829	-0.310578956	-0.3335332676	-0.178960133
8711561	0.099698087	0.4650163199	0.099488405	0.3445297677	-0.210509494
8711803	1.357000000	-0.2047718302	-0.867747813	-0.6261315721	-0.449798994
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8712064	0.162634267	0.0759686030	-0.025276560	0.1336082635	0.137099934
8712289	-0.489780134	0.1825823628	-1.124863782	-0.5377603796	0.116041790
8712291	0.109620313	0.5161052587	0.034472312	0.3811196278	-0.028240701
87127	-0.089845835	0.0284098358	0.287059437	-0.0510103836	0.124995004
8712729	-0.014198536	-0.4133691845	-0.651871424	-0.0775843489	-0.212245786
8712766	0.465146381	0.1401908142	0.286748688	-0.0212027519	-0.220383314
8712853	0.143728624	-0.2563406778	-0.014448411	0.5727701136	0.055215331
87139402	0.051281260	-0.1958633611	-0.116103049	0.0625618790	-0.167715425
87163	-0.333332612	0.1116416150	-0.539899497	-0.0242891297	0.091977503
87164	-0.364413383	0.4868152678	0.532103307	-0.3165469086	-0.318353526
871641	0.211001158	-0.1842548597	-0.165004650	-0.6447695076	-0.099236083
871642	-0.598515985	-0.6850927904	0.808006853	0.1436154067	0.058720958
872113	-0.124944831	0.3664054955	0.845792170	-0.3100083917	-0.101113159
872608	-0.787484776	1.0356294206	-1.261753574	0.4469427053	-0.205643519
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873357	-0.038496417	0.2570638315	0.551936269	-0.2393680598	0.027923468
873586	-0.419867789	-0.2163387111	0.012862546	-0.3492984258	-0.060250651
873592	-0.993120046	0.4602459816	0.475762029	-0.2778000954	0.367602552
873593	0.858346272	-0.2430450902	0.573328596	-0.5008586213	0.286037154
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873885	0.073512430	0.4110661206	0.182394733	-0.0916176368	0.092795108
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874662	-0.590320909	0.3467165180	-0.194244861	0.5081602730	-0.022239934

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875099	-1.643727374	-1.1555118871	0.035642630	-0.8005438352	-0.022724838
875263	-0.114908379	-0.3537747957	-0.544544834	0.1232671318	-0.191064014
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875878	-0.289289560	-0.3395417378	0.069171689	-0.1175130969	-0.064395200
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877159	-0.036030520	-0.3066397032	0.413044666	0.7412864140	-0.025616202
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9012795	-0.776581336	-0.0370483456	0.432908608	-0.5223914992	-0.234149675
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901549	0.023040886	0.3677072988	-0.363493152	0.1904282520	-0.181755847
901836	-0.228523015	-0.7967181894	0.340720994	-0.2688417551	0.306265576
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925292	1.315903353	0.7410877051	-0.442232067	0.1623571454	-0.310794816
925311	-0.569745725	-0.5567515908	0.371393741	0.0059625202	-0.058401266
925622	0.241409376	-0.1284484850	1.010978343	-0.5618115101	-0.980066745
926125	0.747115186	0.5137902382	0.605723215	-0.9204421040	-0.160428761
926424	0.062596180	0.1232334490	-0.051677832	-0.4039342494	-0.652176589
926682	-0.244588350	0.2225567578	-0.192467698	0.0155416535	0.069913471
926954	0.839772720	0.0963880851	0.157279563	0.2854397183	-0.090917772
927241	0.178471102	-0.6968480142	1.224117875	0.2185054033	-0.229389196
92751	0.143967465	-0.1793386428	0.678300647	-1.1696954484	-0.217152272

	PC16	PC17	PC18	PC19	PC20
842302	0.7444607451	-0.2652373958	-5.490796e-01	0.1336499023	0.3452611113
842517	-0.6482383131	-0.0171970719	3.180176e-01	-0.2473469895	-0.1140327429
84300903	0.3248247158	0.1907506408	-8.789759e-02	-0.3922811953	-0.2043524231
84348301	0.0513250859	0.4822096009	-3.584323e-02	-0.0267241008	-0.4643251074
84358402	-0.1524716533	0.1330252645	-1.869779e-02	0.4610302383	0.0654378154
843786	-0.0127056581	0.1967133529	-2.972771e-01	-0.1297264835	-0.0711745335

844359	-0.0279433054	-0.3200829050	1.230170e-01	0.0035239593	-0.0444495512
84458202	-0.3365097518	0.1614215017	-4.908981e-02	0.2196830773	-0.0106331412
844981	0.5070582930	0.2854754597	-2.870772e-01	-0.1822709687	-0.0197682471
84501001	-1.1318257248	0.7584688292	-2.387387e-01	-0.1275685149	0.1779240553
845636	-0.0138706093	0.2849159741	-6.134944e-02	0.1720437193	0.0937289351
84610002	-0.0187810714	0.1611055355	-4.159689e-01	-0.1367793291	0.0566814179
846226	-0.2920952462	0.0531727295	9.339577e-01	0.9728881275	-0.3157956887
846381	-0.1992780260	0.0096995090	9.339247e-02	0.1010674465	0.1967267792
84667401	-0.1618282348	0.1754310229	2.238806e-01	-0.0531840832	0.3639545281
84799002	-0.6382781617	0.1798611136	-1.718454e-01	0.0906929739	-0.2822197817
848406	0.2443739278	-0.0458383708	-2.208124e-01	-0.0808510853	-0.1668401722
84862001	0.2215767406	-0.0781818510	5.021716e-02	0.3243354151	0.1355909907
849014	-0.1393548032	-0.6855739616	-2.705080e-01	-0.4576544799	-0.0261720852
8510426	0.1209221166	-0.1193080949	1.076572e-01	-0.0023196540	0.0627784940
8510653	-0.1439936507	0.2273189096	-9.075712e-02	0.1205914711	0.1354815837
8510824	-0.2156163397	0.0416265799	-4.312105e-02	-0.2850057948	-0.0664739762
8511133	0.0779077052	0.0516753915	-1.351681e-01	-0.2505840594	0.2833238390
851509	-0.1695420732	-0.5795650326	-1.524509e-01	-0.4622425441	0.2441235537
852552	0.2252070093	-0.9343378255	-4.239600e-01	-0.2004324503	0.0316699323
852631	0.5428895558	-0.2315565339	-9.462169e-02	0.1754609443	0.1261898383
852763	0.0290689442	0.4558248771	-1.573099e-01	-0.2028349509	0.3681010830
852781	-0.2633107617	-0.0230320660	5.164379e-02	0.3674136487	0.0390322000
852973	0.3507423311	-0.2024818286	-5.140403e-01	0.1407887067	0.1935393255
853201	0.0540947447	0.0555750576	-6.736093e-02	0.2280003716	-0.2726557691
853401	0.3051862215	-0.1875623435	-1.935028e-01	0.2018598204	0.0776311316
853612	-0.6162327985	0.1233210520	-5.884654e-01	0.1115639655	-0.2376973065
85382601	0.5443556733	-0.1853860373	-2.905819e-01	-0.1327205151	-0.3102765519
854002	-0.6799791601	-0.0524136346	-1.278924e-01	0.3484526710	0.0977244882
854039	-0.2649508034	0.5116666688	-5.148992e-01	-0.0449359045	0.1396499273
854253	0.3963311465	0.3424367694	-2.523528e-01	0.5088120277	-0.0182262570
854268	-0.1353052207	0.1325066667	-1.212405e-01	0.1760177562	-0.0783161394
854941	-0.4865692197	-0.0171701023	9.749344e-02	-0.3466943841	-0.0311932418
855133	-0.5775705797	0.6852634929	-6.832692e-01	1.0176759372	-0.5936297787
855138	-0.2486170983	-0.3725864432	4.074237e-01	-0.2392380729	0.0653398462
855167	0.0858440131	0.0154061997	-1.124179e-01	-0.2063437638	0.1529980089
855563	0.4138240100	-0.1707322382	3.545141e-01	-0.0644683713	0.0680433928
855625	-0.1168940426	-0.5396200806	3.638398e-02	0.1722399585	-0.3325881394
856106	0.2139096807	0.0237139886	-2.109543e-01	0.0982275226	0.0303717372
85638502	0.3840819798	0.2459056676	-1.469466e-01	0.0157458771	-0.0966009077
857010	-0.0516728726	-0.3229228398	-8.174019e-03	0.1334369447	-0.1998283033
85713702	0.4341320342	0.0058211607	8.968932e-02	-0.2077530623	0.0078418173
85715	-0.0343499856	0.1432864234	1.566433e-01	-0.2386347890	-0.2403366743
857155	0.0675974293	-0.1890272279	8.181192e-03	0.2854018034	0.0309365297

857156	0.1742017650	0.0643542582	5.355821e-02	-0.2002754121	0.2156803986
857343	-0.0583889171	-0.0825241914	-1.534190e-01	0.0859253386	-0.0634891959
857373	0.0267599697	0.2211230693	1.393874e-01	0.0785551072	0.0905176134
857374	-0.1451536668	-0.0455372106	7.295476e-02	-0.0932288239	-0.0218792424
857392	0.4610520393	0.0023119228	-2.167759e-02	-0.0116195406	-0.1570539642
857438	-0.0273991255	-0.1756512312	2.032964e-01	-0.0520325983	0.0649106633
85759902	-0.0455436441	-0.0730332412	-1.409306e-01	-0.0752782585	0.0300697515
857637	-0.0136346681	-0.2686278403	-3.472606e-02	-0.2406998187	0.0511072884
857793	0.0076276816	0.4698692126	-4.504667e-01	-0.3869577956	-0.1062390440
857810	-0.2005621445	-0.2045630549	-1.277670e-01	0.2367277317	0.0578713266
858477	0.2370915417	0.0415790617	5.176941e-01	0.1482043271	-0.4564661004
858970	-0.5946048047	-0.5785348079	1.139374e-01	0.0079325268	0.2453309910
858981	-0.1743616455	-0.0787539271	-1.120703e-01	-0.0880295467	-0.2495215707
858986	0.1448061468	-0.4050970648	6.565549e-02	0.2559123023	-0.4289500187
859196	-0.0842293958	-0.3369495728	2.902474e-01	-0.3645193658	-0.0218847216
85922302	0.2644121106	-0.3189750634	1.006240e-01	0.0343908745	-0.1639586428
859283	0.4712897357	0.1030461736	5.156184e-02	-0.0435454461	0.0120385599
859464	0.0720499724	0.2325538744	1.455717e-01	-0.0720682257	0.0790225044
859465	0.2757212862	-0.2805237812	1.404927e-01	0.0372159025	0.0833649877
859471	-1.0626005498	0.1850617241	-1.347993e-01	-0.2668952190	-0.3814350912
859487	-0.1078434849	-0.1246143671	-1.511215e-03	-0.0201171368	0.0939379250
859575	0.0723592874	-0.3383160278	-3.527509e-01	-0.0525835369	-0.0289468771
859711	1.4898419365	-0.3943927763	-4.943103e-01	-0.2226188045	-0.0047807310
859717	-0.7025900371	-0.3233144668	-2.690081e-01	-0.2103448772	0.0834048136
859983	-0.2314069611	-0.1111854157	4.271094e-02	0.0376418646	0.0935975116
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8610404	0.1968371132	-0.2604316743	5.447432e-03	0.2750953606	-0.1701957699
8610629	0.0605687046	-0.1926071856	-3.781177e-01	-0.2349473090	-0.0699485241
8610637	0.2213986609	0.2564393586	-8.694054e-05	-0.0296330599	0.2318213569
8610862	-0.1022416133	-0.4371565117	6.541248e-01	0.6232020130	0.6572255334
8610908	-0.0569375771	0.0933913214	-6.405911e-02	0.1272588554	0.2678163105
861103	-0.1250507711	0.1965686827	-1.543090e-02	0.1643745864	0.0671390460
8611161	0.1507082688	0.0460500757	-8.528589e-02	-0.0424911087	0.1097945421
8611555	0.2951609043	-0.0727220045	3.850636e-01	0.2975862961	-0.1296675824
8611792	0.8991911377	0.0740975033	2.824389e-01	-0.1775032259	-0.7876402180
8612080	0.0541482024	-0.0050063011	8.935649e-02	-0.0258646039	0.0514446563
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86135501	-0.0404448228	-0.0343789127	2.292535e-03	0.1457468888	0.0364090961
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861598	0.0248136798	0.0233371930	-2.172770e-01	-0.1129248347	-0.2643933655
861648	-0.1835520312	-0.0073079751	6.880183e-02	0.0569711434	0.1014561762
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861853	0.3440552885	-0.2273752521	-1.922491e-01	-0.0781472448	0.2033479975
862009	0.0633872491	0.0310541348	-5.257572e-02	0.1152561678	0.1769926531
862028	0.3622171113	-0.3077571762	-1.249274e-01	-0.0092797913	-0.0162603868
86208	-0.1499599397	0.1836067507	-1.656348e-01	0.0902312461	0.0175660697
86211	-0.6014543270	-0.0610520777	4.088117e-02	-0.1833302668	-0.0604185139
862261	-0.0996387794	0.1457218445	-4.086137e-01	-0.0517074621	-0.0903242792
862485	0.2297949692	-0.1010915701	3.228795e-01	0.0795964606	-0.2354725917
862548	0.0263312435	0.0738390906	2.193437e-01	-0.0561248044	0.0934256201
862717	0.2603574888	-0.0843126026	-3.760217e-01	-0.1282177279	0.1466945932
862722	-0.0648161007	-0.0743586672	6.001312e-02	-0.1605585403	-0.1381556177
862965	0.3166072907	0.1782153925	1.222033e-01	-0.1914123729	0.1774645649
862980	0.1904717597	-0.2912688556	3.675219e-01	-0.1108806079	0.1781375653
862989	-0.3723552575	-0.2544929562	7.295163e-02	-0.0412255879	0.0550120885
863030	-0.3423285309	0.0370571626	-3.073614e-01	-0.1802702994	-0.1095713573
863031	0.0260291585	0.0416980798	1.873460e-01	0.0537284809	0.0987456977
863270	0.3955030905	0.3233273965	2.988237e-02	-0.0777973821	0.0970946107
86355	0.4535034431	-0.3193808862	-3.373791e-02	-0.1463574077	-0.6693950389
864018	0.2303559150	-0.2079807983	5.509481e-01	0.3335341685	-0.3083514381
864033	-0.1147786362	-0.2432508051	-1.125072e-01	-0.0451754187	-0.0888126673
86408	-0.0914461903	0.2412463437	-1.459792e-01	-0.1100026847	-0.0896473479
86409	0.5978782675	-0.1822196905	-2.997565e-02	0.3190458027	0.4277165587
864292	-0.3616205824	0.2327580071	-2.695597e-01	-0.1740128225	0.0009258643
864496	-0.1170660795	0.1141917260	2.787614e-01	-0.2699318125	-0.4862268924
864685	0.0181444217	-0.1943937215	2.229841e-01	0.1106013071	-0.1943231201
864726	0.0188024294	0.3540363209	-5.535981e-01	-0.1880082735	-0.1775926041
864729	0.3784407497	-0.3282198389	2.854566e-01	0.1676563450	0.1316948737
864877	-0.3174913261	-0.3551554039	1.613097e-01	0.0770344109	-0.3060738877
865128	0.0741606002	0.1293691405	-9.533673e-02	0.1598354031	0.1444500222
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86517	-0.1508543945	-0.0571332361	2.183625e-01	0.0936904685	-0.0794546001
865423	0.8654238320	0.7210245200	-2.842764e-02	-0.1805240610	-0.1953916687
865432	-0.0786214283	0.1112746213	-2.389655e-02	0.0003689045	0.0938018536
865468	0.1787431134	-0.0528239193	9.687076e-02	-0.1321926180	0.0656264988
86561	-0.2137144308	0.0204290791	4.576725e-02	0.1421019311	0.0844834024
866083	0.1945433447	-0.0616013058	1.201657e-02	0.0727533344	-0.0556195344
866203	-0.4075855406	-0.1063572544	-4.203945e-02	0.1864270776	-0.0899024211
866458	0.1487266791	0.0365407567	7.118367e-02	-0.0460559372	0.2878951781
866674	0.1997560730	0.1643581615	-1.386212e-01	-0.2355432488	-0.0786941556
866714	-0.0321454015	0.1803153879	1.446663e-02	0.0537012203	-0.0672121974
8670	0.2690891739	-0.4026752272	-1.168613e-01	0.0558655115	0.0581974748
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867387	0.1255354782	0.0181058265	5.293830e-02	-0.0645255704	0.1452601843
867739	0.0498471273	-0.0348003591	-1.552902e-03	0.3088338323	-0.1359873769

868202	0.1514154715	0.1551834702	1.691757e-01	-0.0013782678	-0.0002043019
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868682	0.3202988718	0.1431577163	-1.510629e-01	-0.3503780602	0.0845967628
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869254	0.4046408528	-0.0226426149	-1.825173e-01	0.0246703885	0.0416271894
869476	-0.2809777491	-0.3045726592	1.129379e-01	0.0215657449	-0.1412924695
869691	0.3205850216	0.2717847437	3.110889e-02	-0.4585564331	0.3093448681
86973701	-0.0076659266	-0.0866896819	1.849691e-01	0.1761969789	0.1865667960
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8711002	0.2108569576	0.1876319970	-1.022811e-02	0.2493521245	-0.1143566685
8711003	-0.0111404858	0.0965996634	1.425451e-01	-0.1091326190	-0.0610483741
8711202	0.3261857915	-0.0289408419	-7.767180e-02	0.0913939848	-0.1968742098
8711216	-0.2234369779	0.2057459455	7.729543e-02	0.2132831815	-0.1182213703
871122	0.0960065509	0.0493671338	6.776082e-02	-0.0892318905	0.0038008994
871149	0.5371407724	0.0836013519	-7.347705e-02	0.2296796112	-0.0223156244
8711561	-0.0172732266	-0.1660518913	-2.693205e-01	0.2789556075	0.2686127044
8711803	-0.1739718477	-0.1186035199	1.495407e-01	0.1184277556	0.2759074386
871201	0.1985593591	-0.3130446110	-7.422400e-01	-0.4027477710	0.0294493406
8712064	-0.1408167763	0.0899760234	4.338600e-02	0.0516195709	-0.0014061964
8712289	-0.5541220604	0.4209667792	1.441757e-01	-0.2427057037	-0.1668334484
8712291	-0.0270993383	0.1694953530	-8.852552e-03	0.0891065240	0.1385092890
87127	0.1403572442	-0.1271309456	1.672506e-01	0.0484315494	-0.1564043597
8712729	0.0344464249	-0.1026197113	4.597199e-04	-0.1001347432	0.0698983002
8712766	0.3396355704	-0.3172804519	-5.565185e-01	0.2086270672	-0.1057423397
8712853	-0.0661256124	0.0947245242	5.850816e-02	0.0267445695	0.0761671148
87139402	0.0391426633	-0.0894043810	4.249078e-02	-0.1104277441	0.0347903302
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875099	-0.5209914969	-0.6680161184	4.027139e-01	-0.2888446698	-0.6022166887
875263	-0.2098332786	-0.1954162464	4.234095e-01	0.1016542829	0.0172641121
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91903902	-0.0589653404	0.1378144328	-5.111800e-02	0.1123664221	-0.0998720750
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919537	0.1671650757	0.1637494512	2.267327e-01	-0.2011274932	0.0996771994
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91979701	0.1441660514	0.0503790828	7.972177e-02	-0.0689864988	0.1948700448
919812	-0.1845235859	0.2560471166	2.157609e-01	0.0185852693	0.1565496905
921092	0.2246393773	-0.0759854108	-2.144004e-01	0.1077679119	-0.0754591770
921362	0.1157153017	-0.0946085126	5.607015e-01	-0.1632070601	-0.4108193773
921385	-0.0562823823	0.1692280446	-1.213046e-02	-0.0535534599	0.1078587734
921386	-0.1524138427	-0.0391271728	1.677550e-01	0.1610284957	-0.1314182663
921644	-0.2181697442	0.0270419575	7.676963e-02	-0.0335181090	0.1858465398
922296	0.1223984599	0.1891922492	-3.446057e-02	-0.2263650827	0.0703565400
922297	-0.4367745068	0.0792966644	1.675768e-01	0.2307348371	-0.0359430593
922576	-0.0494701048	0.0013069407	4.558326e-02	-0.0629478688	-0.1050644118
922577	0.0091468550	0.0290324891	1.488410e-02	0.0181300769	0.0288337616
922840	0.2476202326	-0.1276536694	4.694131e-01	-0.2003872236	-0.0195139777
923169	0.1388873508	-0.0515030353	-4.852651e-02	0.0264827985	0.1271702623
923465	0.0847374464	-0.1500183423	-7.870783e-02	0.3579268358	-0.1824970475
923748	-0.0103802678	-0.3204581803	-9.803317e-02	0.0202986827	0.0295470955
923780	-0.0057459625	-0.0405914791	6.431842e-02	-0.3217160263	-0.1685804913
924084	-0.0573291350	0.0452087235	1.212636e-01	-0.0765829801	-0.0502932900
924342	-0.4351182523	-0.1917622682	-1.571844e-01	-0.2947566845	0.1635577852
924632	0.0048691691	-0.0303433227	8.876475e-02	0.0041081275	-0.1145334139
924934	0.0206805941	0.1334742843	1.145619e-01	-0.1800354098	0.1188747634
924964	-0.2612530693	-0.0405031771	-1.773446e-01	-0.2615734828	-0.0134633024
925236	0.1100555556	-0.2497956869	-4.510752e-01	0.2787558115	-0.0588049466
925277	-0.1194566710	0.1244195792	1.484120e-02	0.0607378108	0.0576588492
925291	0.2761350539	0.2273597903	-6.871634e-02	-0.0276269369	0.1478646421
925292	-0.1693243274	0.0815532304	-2.099437e-02	0.0818645399	0.2514566080
925311	0.1807754237	0.0762688515	-3.325661e-01	0.0845524505	-0.0415106978
925622	-0.8029651181	0.1220605895	-4.950476e-01	-0.1819617089	0.3143066179
926125	0.0098729574	-0.0800009784	6.354433e-02	0.5562492156	-0.1554353516
926424	0.1475121301	0.3479599809	-1.950425e-01	-0.2460983953	-0.4040909029

926682	-0.1380625998	0.2932365347	-1.089496e-01	-0.1823602477	-0.2297445474
926954	-0.2324439379	-0.0655571760	2.108950e-02	-0.0419829133	-0.0813483422
927241	-0.0609931700	-0.1683659911	-3.066037e-01	0.3102957595	0.1730640918
92751	0.9204776932	-0.3045424620	-1.808133e-01	0.1889373346	0.1631105493
	PC21	PC22	PC23	PC24	PC25
842302	0.0964300446	-0.0687893946	8.444429e-02	0.1751022126	0.1508872943
842517	-0.0772594939	0.0944953050	-2.175267e-01	-0.0112801927	0.1703603547
84300903	0.3107932463	0.0602560125	-7.422581e-02	-0.1026714190	-0.1710076559
84348301	0.4338116608	0.2030870644	-1.239955e-01	-0.1532947800	-0.0774275740
84358402	-0.1164424689	0.0176343316	1.393310e-01	0.0053271101	-0.0030593705
843786	-0.0024001785	0.1010804263	3.344819e-02	-0.0028377493	-0.1222827647
844359	-0.0497039122	-0.0909661524	-1.428548e-01	0.0801136391	0.2774169346
84458202	0.0446160665	0.1312420402	-2.402675e-01	-0.0962489988	0.1953110200
844981	0.0956161490	-0.1961497958	-5.867294e-02	0.0884762672	-0.1302274924
84501001	-0.2564517862	0.7160864577	1.671813e-01	-0.2158422553	-0.1346590354
845636	-0.1931514370	0.1797289643	1.239130e-04	-0.0661301394	0.1793508578
84610002	0.0256613371	0.1785045463	-2.105003e-01	0.1707930257	-0.2775786245
846226	-0.0212336227	-0.2717732995	8.302672e-01	0.5694022242	0.2050807032
846381	0.0536465611	0.2250613132	1.057031e-01	-0.0118578557	0.0109923447
84667401	0.1057076291	0.0548565083	5.742781e-03	0.1230654885	-0.1363435633
84799002	-0.0751373662	0.0433461466	1.334490e-01	0.1175438173	-0.2497804016
848406	-0.2344824832	-0.0800755365	-1.157460e-01	-0.0257880551	0.1982873357
84862001	-0.1000151185	0.2316735867	-1.000205e-01	-0.1851440561	0.2205773586
849014	-0.0712281618	-0.3695846503	2.874512e-02	0.3278945116	0.1650638434
8510426	-0.0042437600	-0.0571347487	2.945411e-02	0.0927005896	0.0899440626
8510653	0.1108892757	-0.1668964790	3.887662e-03	-0.0412016079	-0.0692929154
8510824	-0.0733857126	-0.0551410551	1.379913e-01	-0.0349132014	-0.2320428771
8511133	0.1246145246	-0.2051975642	-7.854400e-02	-0.0946139362	-0.0752040336
851509	-0.1903107396	0.1253718436	-1.521652e-01	0.1059576635	0.0372312067
852552	0.1055683715	0.1636112461	1.458986e-01	0.1809515336	0.2961245192
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852763	0.0163246146	-0.0141963492	-5.970130e-01	0.2221582674	0.0103514077
852781	-0.1014472410	0.0490161400	-3.782756e-02	-0.2044090783	-0.0458583192
852973	0.0617522965	-0.2965392069	5.075293e-02	0.2219116625	-0.3516296603
853201	-0.1913776215	0.0214862498	-2.929037e-01	0.1208650268	-0.1369103839
853401	0.0595587515	0.2771629842	1.088098e-01	-0.0532229335	0.1666814036
853612	-0.1063311988	-0.2066568837	2.611420e-01	-0.1086780448	0.0057569554
85382601	-0.0434313811	-0.0112246042	5.029196e-02	0.0301623182	-0.1101463645
854002	0.6876197560	-0.0004074532	9.408438e-02	0.0229190936	-0.3765055802
854039	0.0504027758	0.2876283914	-4.554592e-02	0.0539289063	-0.0656662192
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854941	0.2840168766	0.0526557024	1.154075e-01	0.0203929562	-0.1382255860

855133	0.4043981179	-0.1406766632	1.049883e-01	-0.3703681389	-0.3094285067
855138	0.2089804524	-0.2310395766	-4.420531e-01	0.1221456978	0.2819915427
855167	-0.1590984393	-0.1873591123	-2.090009e-01	-0.0034415007	0.0283145970
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856106	0.1630859956	0.1050367727	3.885715e-03	-0.0553808770	0.1268753135
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857343	0.0627384666	0.0519012423	1.971663e-02	-0.0571482290	-0.0566535809
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858981	0.1228232989	0.0935845279	8.347651e-02	0.1223396053	0.0212412139
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859471	-0.0179672286	0.1239548335	-5.671419e-01	-0.1119284311	0.2249337178
859487	0.0321781929	0.0892493026	9.990179e-02	0.0548699108	-0.0126058236
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859983	0.0737021390	0.2859701894	-2.018351e-01	0.0772917781	0.1152392295
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868826	-0.3334506884	0.4054915252	-2.193496e-01	-0.4068775025	-0.0881771690
868871	0.2803794460	0.1846761587	2.143952e-01	-0.1196863856	0.0376078871
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869224	0.1732044592	-0.0639742623	1.991783e-02	0.0106138462	0.0122799552
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869476	-0.3683875482	-0.1420416676	-1.308197e-01	0.0125169124	-0.0452869192
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87106	0.0037288783	-0.0076158198	4.975953e-02	0.0287897188	-0.0665129766
8711002	0.1329682058	-0.0117514493	3.214991e-01	-0.0543627264	0.0702960637
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8711202	0.1130023897	0.2175450326	1.330678e-04	-0.1083858543	-0.1131675407
8711216	-0.1196078069	-0.1293049567	-1.269144e-01	-0.1667389417	0.0522691793
871122	-0.1124828396	0.1751555824	-6.726155e-03	0.0674615200	0.0668955418
871149	0.0731062518	0.2607303508	8.887002e-02	0.0439906279	0.0851068271
8711561	-0.0593151545	-0.0152866466	-9.779063e-02	0.2113824287	0.1635538400
8711803	-0.1263678204	0.3605520731	-2.127513e-02	-0.1363983763	-0.0898596892
871201	0.0478377893	-0.0017237997	4.327957e-01	-0.0776408430	0.1636625284
8712064	0.0725704198	-0.0013301824	-1.161848e-01	-0.0247003946	0.0586874936
8712289	0.1620788140	-0.1216703596	-1.509285e-01	0.1083602709	-0.0275326905
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87127	0.0818090553	0.0769269570	9.751749e-02	0.0094024462	-0.1095299583

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871641	0.0090184455	-0.1279528819	2.305423e-01	-0.0197482542	-0.1580653111
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895299	0.0118916661	0.0576633586	4.487270e-03	-0.0790272613	-0.0632766829
8953902	-0.2017386600	0.0416237508	1.260261e-01	0.0585570939	-0.1012969876
895633	-0.0307844139	0.1771917416	-9.179344e-02	-0.2293132698	-0.0364655954
896839	0.0675806388	-0.1599562906	-1.802517e-01	0.0134278792	-0.0158110877
896864	0.1584405886	-0.2240196469	2.841299e-02	0.1527066450	0.1002162969
897132	0.0554024832	-0.0106718651	3.772549e-02	0.0739356388	0.0595961420
897137	-0.1804721118	0.1863579187	1.526429e-01	0.0270508295	-0.0952060171
897374	-0.1722866586	0.1118078089	1.027824e-01	-0.0579933331	-0.0718227119
89742801	-0.4406938771	-0.1185551117	-1.021070e-01	0.0808540663	-0.0388498952
897604	-0.0166058998	0.0753598960	-1.673156e-01	-0.1454966521	0.0884728655
897630	-0.0426489588	-0.1276660353	-2.540048e-02	-0.1547491044	-0.0398592453
897880	0.0496498233	-0.0014560954	-6.804934e-03	0.0546181438	0.0559208755

89812	0.1260013993	0.2302910268	2.697330e-01	0.0986525411	-0.1072451982
89813	-0.0513658932	-0.2706243346	-1.825089e-01	-0.0269983135	0.0260255900
898143	-0.1400771426	-0.1592083408	-5.743311e-03	-0.0971432330	-0.2176794045
89827	0.0004474165	-0.0950021327	-7.854093e-02	0.0749693728	0.0697136045
898431	-0.4411488718	-0.1066038359	-9.846925e-02	0.1435810774	0.0111032267
89864002	-0.1119577804	-0.0689397196	1.201122e-02	0.0938932586	-0.0037422331
898677	0.0082550834	-0.0635324658	4.878671e-02	-0.0603886508	0.1389374318
898678	-0.0383648376	0.1969142382	4.517753e-02	-0.0303553905	0.0523315363
89869	0.0279241117	-0.0786297359	-7.174673e-02	0.0350504919	0.1850126955
898690	0.1278967639	0.2146406685	1.210090e-01	0.0271655527	0.1154009068
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899187	0.0275592917	0.0537676277	1.451095e-02	0.0082236013	0.0009948128
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899987	0.2886257323	0.2696377063	2.790016e-01	0.3620797422	-0.4238626098
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901011	-0.1360964486	0.1906190355	1.897326e-01	0.0084397063	-0.1152481953
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9010333	-0.2408557256	0.1082122384	2.815897e-02	0.0807684948	-0.1187718156
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901034302	0.0346629395	0.0850128176	3.488674e-02	0.0356471287	-0.0344347567
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9010877	0.1412840554	0.0694193070	-5.871105e-03	0.0145527182	0.0389700365
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9012568	0.0593266847	0.0445037552	7.089414e-03	0.0025516233	-0.0597606582
9012795	-0.2379071139	0.0909867771	-1.393259e-01	-0.0365566691	-0.0293749486
901288	0.1371863492	-0.2443319223	-1.319252e-01	0.0115020589	-0.1567813559
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9113538 -0.1457441362 7.226593e-02 -5.951642e-02 1.311257e-02 -3.413340e-
02
911366 0.0957538784 -3.485376e-02 2.750362e-02 -1.043926e-02 -6.172565e-
06
9113778 0.0909946383 1.398598e-01 8.858450e-03 7.244742e-04 -1.250954e-
02
9113816 0.0730814567 6.023963e-02 1.322804e-02 5.220342e-03 -9.310005e-
04
911384 0.0226017652 -1.609558e-02 2.644659e-02 -2.709195e-02 -6.915033e-
03
9113846 -0.1242840374 -5.698309e-02 3.319390e-03 5.876624e-03 6.378471e-
03
911391 0.0182447696 4.114275e-02 -1.661732e-02 -7.879829e-03 4.976152e-
04
911408 0.1375631124 -1.253250e-01 6.518456e-04 -1.097810e-03 -4.923167e-
03
911654 -0.0829225760 7.105777e-02 -4.770183e-02 -6.996362e-03 -9.172107e-
04
911673 -0.0827292480 -3.235318e-02 -1.469619e-02 7.414257e-03 5.606623e-
03
911685 0.0448464007 4.121335e-02 -2.359053e-02 2.794497e-02 -3.572364e-
03
911916 -0.0495987508 -2.268418e-02 2.227878e-02 8.961678e-03 -8.038788e-
03
912193 -0.0128831723 -1.410240e-02 -2.422196e-02 2.001356e-02 -1.799201e-
03
91227 -0.0381183878 2.857374e-02 1.498774e-03 -2.545113e-02 -5.760572e-
03
912519 0.0403505601 -2.930731e-02 4.188617e-02 1.191858e-02 3.348884e-
03
912558 -0.0154989113 8.817212e-03 9.226792e-03 1.934553e-03 1.020699e-
02
912600 -0.0136577142 -6.945356e-02 7.542745e-03 -1.162578e-02 1.376873e-
02
913063 -0.0107779990 3.093484e-03 1.413849e-01 -6.100949e-03 1.202071e-
03
913102 0.0095431487 -7.376165e-03 2.000637e-02 -2.068873e-02 -5.432457e-

03
913505 0.0040869238 1.517483e-02 1.015079e-01 -1.262707e-02 3.374017e-
03
913512 0.0355692398 9.085520e-02 -4.393050e-03 4.668312e-03 -6.227434e-
03
913535 0.0086380352 1.069920e-02 -3.724756e-02 -2.337908e-02 9.673922e-
03
91376701 -0.0523983166 -2.155044e-02 -3.015589e-02 4.206849e-03 -4.615799e-
03
91376702 0.0364243509 2.194271e-02 6.585056e-03 -5.260954e-03 1.175358e-
02
914062 -0.0249186250 -9.840999e-02 -1.409311e-02 -4.124637e-03 -2.145734e-
03
914101 -0.0546115161 -1.086523e-01 2.580461e-02 6.505626e-03 3.807495e-
03
914102 -0.0900970424 -7.973081e-02 -1.382759e-02 2.391790e-02 2.493993e-
03
914333 -0.0040977809 -6.976257e-02 -1.359450e-02 -7.869232e-03 4.008083e-
03
914366 0.0965211814 -5.682683e-02 6.498587e-03 1.456788e-02 2.151391e-
03
914580 -0.0422754385 2.085169e-02 -1.607462e-03 3.798926e-03 -1.110263e-
02
914769 -0.0967977019 -1.485310e-01 1.576602e-02 3.329348e-03 -9.341349e-
03
91485 -0.0641045051 1.841422e-01 -7.024287e-02 1.970595e-02 -1.498247e-
02
914862 0.0854465362 -1.747535e-01 -3.607397e-02 1.867275e-03 -3.821167e-
03
91504 -0.0963197757 9.104160e-02 -2.312398e-02 3.386630e-02 -1.529871e-
02
91505 -0.0121822447 -1.090114e-02 8.724908e-03 4.694641e-03 -1.074887e-
03
915143 -0.1180678238 2.658401e-01 -7.894180e-02 -2.561913e-02 7.697159e-
03
915186 0.1148367203 2.121193e-01 -3.514142e-02 -5.061458e-02 3.668640e-
02
915276 0.2752439895 -3.019773e-01 9.673065e-03 -1.996698e-02 4.151417e-
02
91544001 -0.0485533966 5.416690e-02 -1.649414e-02 -1.490646e-02 -2.155321e-
03
91544002 -0.0036376007 -1.802591e-01 -1.011343e-02 2.070820e-02 6.669538e-
03

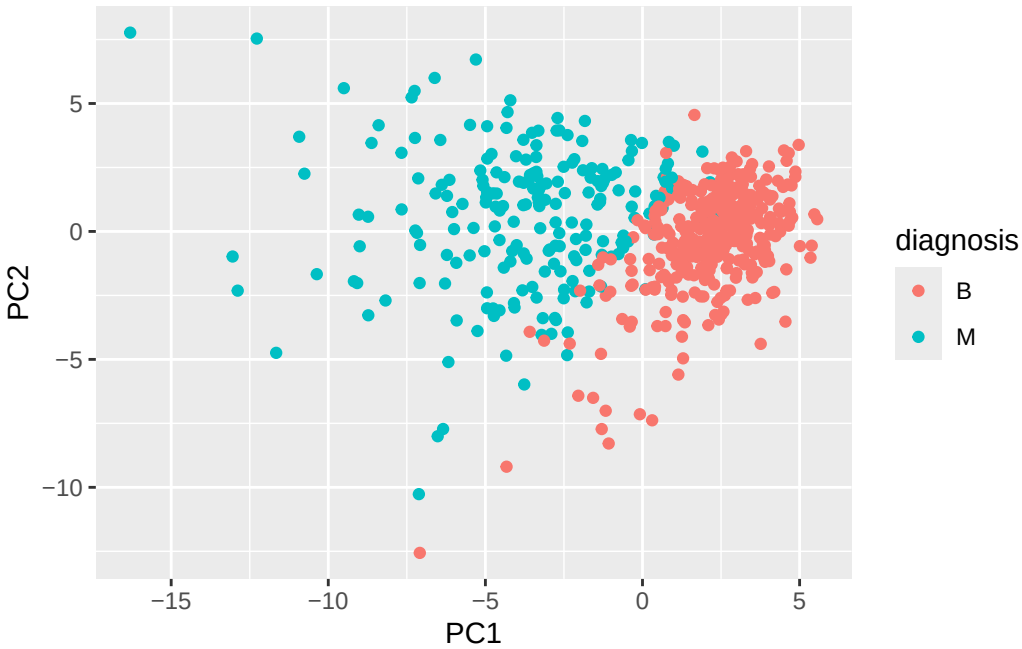
915452 -0.0647665350 4.842798e-03 1.594580e-02 9.613952e-03 -3.100001e-
04
915460 -0.1754187549 1.359165e-01 4.067320e-02 5.041105e-02 2.144200e-
04
91550 -0.0556837067 6.632740e-02 -6.746846e-03 1.531456e-02 -1.218153e-
02
915664 -0.0984427406 1.808916e-02 1.255164e-02 9.930751e-03 9.221713e-
03
915691 0.0036057142 -8.355934e-02 -3.592810e-02 2.235170e-02 3.151072e-
03
915940 -0.0112187011 2.530924e-02 4.763963e-03 -1.126705e-02 7.608210e-
04
91594602 -0.0091762716 -7.293476e-02 2.792662e-02 -3.556928e-03 -1.164186e-
03
916221 0.0038142602 3.300541e-02 8.325671e-03 -4.509973e-03 -1.285385e-
03
916799 0.0189855035 -9.752934e-02 -6.505096e-03 -2.312757e-03 -1.589520e-
02
916838 0.0550042618 -1.724436e-01 -9.894992e-02 -7.051032e-02 7.208234e-
03
917062 0.1580673727 -6.410352e-02 -8.250866e-03 -1.932592e-02 1.086353e-
02
917080 0.0544002611 1.580256e-02 -1.613705e-03 8.423291e-04 9.014139e-
04
917092 0.0074556575 9.663785e-02 -2.665164e-02 -3.072198e-02 -5.603826e-
03
91762702 0.2131860543 1.741659e-01 -1.264691e-01 1.220225e-02 -8.726745e-
03
91789 -0.0681589606 -9.361363e-03 -8.419239e-03 1.344440e-02 2.051314e-
03
917896 -0.1061345671 4.966193e-03 -6.582652e-03 4.193781e-02 7.898565e-
03
917897 0.0664531179 -2.655619e-02 -2.136312e-02 -9.637608e-03 2.497448e-
03
91805 0.0886147194 1.155966e-01 -5.229288e-02 1.076846e-02 1.449954e-
03
91813701 0.0330021360 8.336065e-02 -5.061548e-03 3.298927e-02 5.387830e-
04
91813702 0.0181336410 4.282084e-02 3.040698e-02 1.508284e-02 -7.330015e-
03
918192 -0.0934720894 -5.020254e-02 4.171083e-02 -7.809330e-03 -9.193665e-
05
918465 0.0288790490 -5.652693e-02 -1.947227e-03 1.611639e-03 -4.255828e-

03
91858 0.0424242869 9.025177e-03 -1.623234e-02 6.269355e-03 -8.871060e-
04
91903901 0.0931826121 -3.654002e-02 4.324270e-03 4.165806e-03 -1.106227e-
03
91903902 -0.0830740552 1.658681e-02 1.150422e-02 -1.563064e-02 -6.663151e-
03
91930402 0.1480707171 8.699232e-03 -5.961362e-02 -1.044751e-02 -6.141284e-
03
919537 0.0728753384 1.259825e-02 -9.518713e-03 9.373100e-03 -3.095154e-
03
919555 0.1536070303 -3.265112e-02 2.996697e-02 8.171708e-03 -3.458037e-
02
91979701 -0.0487762627 5.821042e-02 1.015633e-02 2.035367e-02 9.738559e-
03
919812 0.0329233687 -1.804282e-02 -2.169062e-02 -2.378939e-02 8.893677e-
03
921092 0.1468627182 7.253893e-02 -8.746899e-03 -1.161071e-02 9.286171e-
03
921362 0.1812441331 7.132198e-03 1.665571e-03 -5.739799e-02 3.679201e-
02
921385 -0.0295036155 -8.523727e-03 -8.796157e-03 -2.463757e-03 9.112497e-
05
921386 -0.0987152511 2.575798e-03 -1.994940e-02 8.161193e-03 -1.262046e-
02
921644 -0.0565374950 1.610918e-02 -1.295072e-02 -2.390649e-02 4.092165e-
03
922296 0.0568923371 -7.840121e-02 -1.162042e-02 3.505356e-03 -6.121733e-
03
922297 -0.0542291424 -8.247547e-02 -8.960736e-03 1.302883e-02 4.772981e-
03
922576 -0.0678745818 -1.961314e-02 -6.487119e-03 -4.750888e-03 -3.114620e-
03
922577 0.0125381679 4.498783e-02 5.470187e-03 -7.783998e-03 -4.507372e-
03
922840 0.0313546289 -4.161718e-02 -2.705643e-03 9.706143e-03 3.657914e-
03
923169 0.1006525985 5.511332e-02 -8.274135e-04 -2.590615e-02 8.789459e-
04
923465 -0.0007909619 3.418774e-02 6.937288e-05 -2.145122e-02 -4.134624e-
03
923748 -0.0163685109 -1.107701e-02 2.770954e-02 1.795767e-02 8.260042e-
03

```
923780    -0.0014215153  5.637368e-02 -3.381745e-03  3.849594e-03 -1.779362e-
03
924084    -0.1022492547 -9.323360e-03  5.913651e-04  1.263059e-02 -7.162542e-
03
924342     0.0044220903  4.297202e-03 -9.650436e-03 -1.615592e-02 -1.435061e-
03
924632     0.0024994349 -7.130661e-02  4.473723e-02  1.560585e-02  1.385123e-
03
924934     0.0938320141  3.896512e-02 -1.164274e-02 -1.431460e-03  6.878081e-
03
924964     0.0152530115 -3.226473e-03  2.268760e-03 -1.799952e-02 -4.947889e-
05
925236     0.1510723322  1.702667e-01  3.413466e-02  3.102528e-02  8.728220e-
03
925277     0.0407493494  3.978350e-02 -1.130892e-02  4.068695e-03 -6.920526e-
03
925291     0.0599659983 -3.787753e-02  2.037519e-02 -3.413149e-04  1.269775e-
02
925292     0.0968312180 -1.392626e-02  2.900866e-02  1.731854e-03  4.328273e-
03
925311     0.0041827618  2.209661e-02  7.486086e-03 -3.107530e-04  5.831723e-
03
925622     0.2037854413 -1.474368e-01 -1.245986e-01  9.015380e-03 -7.118296e-
03
926125    -0.0729289516  1.450989e-01 -3.945551e-02  7.014761e-02  7.638391e-
03
926424    -0.1227922446  1.590577e-01  9.843968e-03  7.112909e-03  1.413613e-
03
926682     0.0658312978 -6.845211e-02 -2.355466e-02 -2.746212e-02  1.041027e-
02
926954     0.0070890278 -2.367063e-02 -2.575490e-03  6.700316e-03 -3.767102e-
03
927241    -0.1980178555 -2.738532e-02 -6.802653e-02 -8.408929e-02  2.376191e-
02
92751      0.3468098821  1.272691e-01 -9.507513e-03 -6.085956e-02  1.945846e-
02
```

```r
library(ggplot2)

ggplot(wisc.pr$x) +
  aes(PC1, PC2, col = diagnosis) +
  geom_point()
```




```r
summary(wisc.pr)
```

```
Importance of components:
                          PC1    PC2     PC3     PC4     PC5     PC6     PC7
Standard deviation     3.6444 2.3857 1.67867 1.40735 1.28403 1.09880 0.82172
Proportion of Variance 0.4427 0.1897 0.09393 0.06602 0.05496 0.04025 0.02251
Cumulative Proportion  0.4427 0.6324 0.72636 0.79239 0.84734 0.88759 0.91010
                           PC8    PC9    PC10   PC11    PC12    PC13    PC14
Standard deviation     0.69037 0.6457 0.59219 0.5421 0.51104 0.49128 0.39624
Proportion of Variance 0.01589 0.0139 0.01169 0.0098 0.00871 0.00805 0.00523
Cumulative Proportion  0.92598 0.9399 0.95157 0.9614 0.97007 0.97812 0.98335
                          PC15    PC16    PC17    PC18    PC19    PC20   PC21
Standard deviation     0.30681 0.28260 0.24372 0.22939 0.22244 0.17652 0.1731
Proportion of Variance 0.00314 0.00266 0.00198 0.00175 0.00165 0.00104 0.0010
Cumulative Proportion  0.98649 0.98915 0.99113 0.99288 0.99453 0.99557 0.9966
                          PC22    PC23   PC24    PC25    PC26    PC27    PC28
Standard deviation     0.16565 0.15602 0.1344 0.12442 0.09043 0.08307 0.03987
Proportion of Variance 0.00091 0.00081 0.0006 0.00052 0.00027 0.00023 0.00005
Cumulative Proportion  0.99749 0.99830 0.9989 0.99942 0.99969 0.99992 0.99997
                          PC29    PC30
Standard deviation     0.02736 0.01153
Proportion of Variance 0.00002 0.00000
Cumulative Proportion  1.00000 1.00000
```

> Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

0.4427

> Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

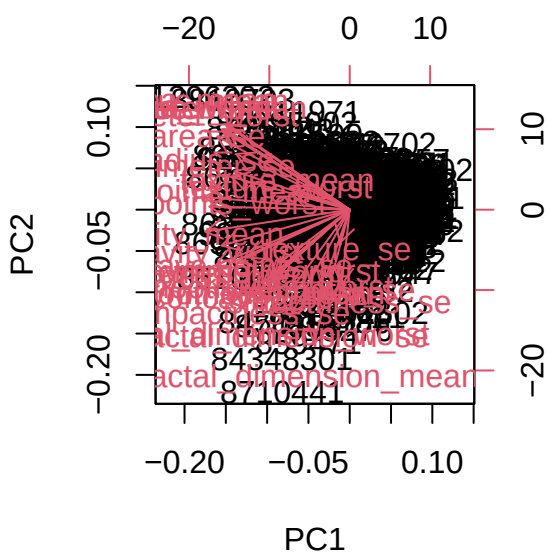
There need to at least be three PCS to describe 70% of the original variance.

> Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

There are 7 PCS required to describe 90% of the original variance in the data.

Interpreting PCA Results

```
biplot(wisc.pr)
```



> Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

It is difficult to understand because distinctions cannot be easily made.

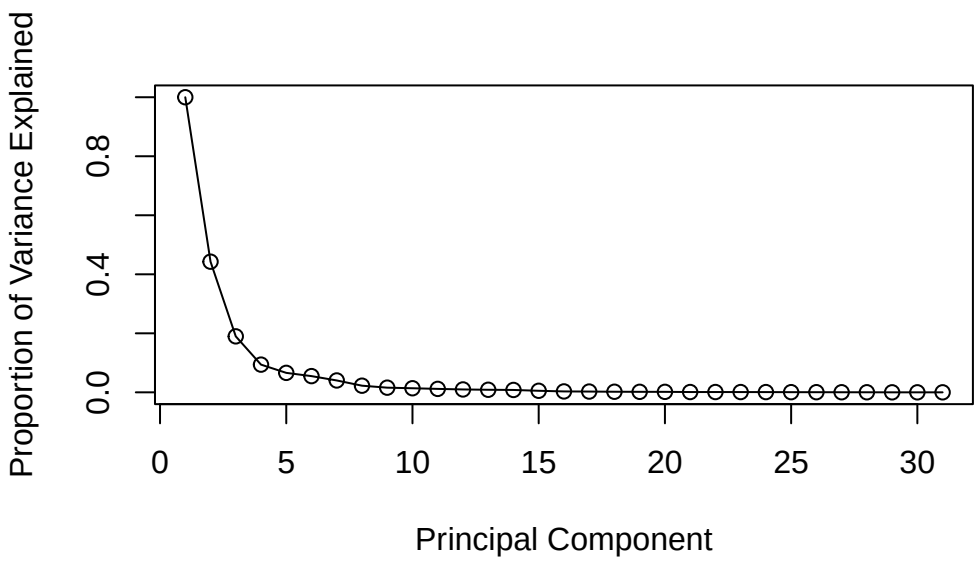
Variance Explained

```
pr.var <- wisc.pr$sdev^2
head(pr.var)
```

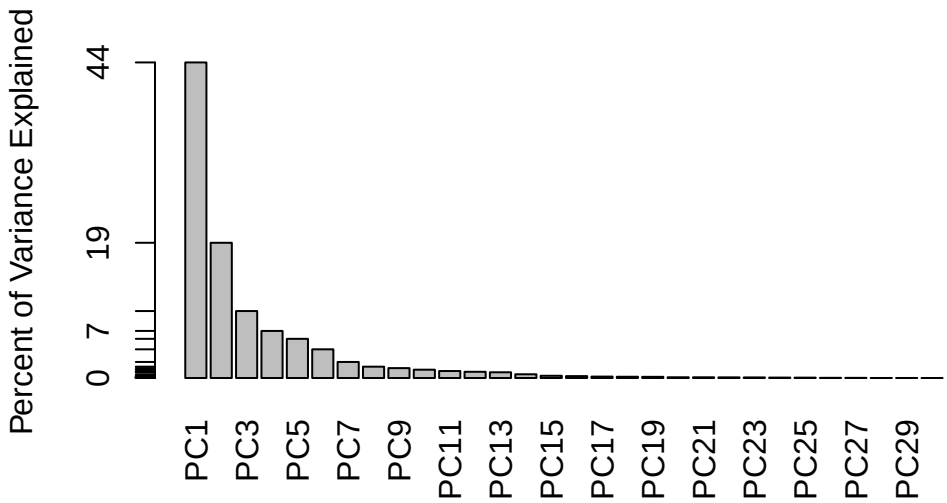
```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

```
pve <- pr.var / sum(pr.var)

plot(c(1,pve), xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1), type = "o")
```




```
barplot(pve, ylab = "Percent of Variance Explained",
     names.arg=paste0("PC",1:length(pve)), las=2, axes = FALSE)
axis(2, at=pve, labels=round(pve,2)*100 )
```



Q9. For the first principal component, what is the component of the loading vector (i.e. wisc.pr$rotation[,1]) for the feature concave.points_mean? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

```
wisc.pr$rotation[,1]
```

```
            radius_mean           texture_mean         perimeter_mean
            -0.21890244            -0.10372458            -0.22753729
              area_mean        smoothness_mean       compactness_mean
            -0.22099499            -0.14258969            -0.23928535
         concavity_mean    concave.points_mean          symmetry_mean
            -0.25840048            -0.26085376            -0.13816696
 fractal_dimension_mean              radius_se             texture_se
            -0.06436335            -0.20597878            -0.01742803
           perimeter_se                area_se          smoothness_se
            -0.21132592            -0.20286964            -0.01453145
         compactness_se           concavity_se      concave.points_se
            -0.17039345            -0.15358979            -0.18341740
            symmetry_se   fractal_dimension_se            radius_worst
            -0.04249842            -0.10256832            -0.22799663
          texture_worst        perimeter_worst              area_worst
```

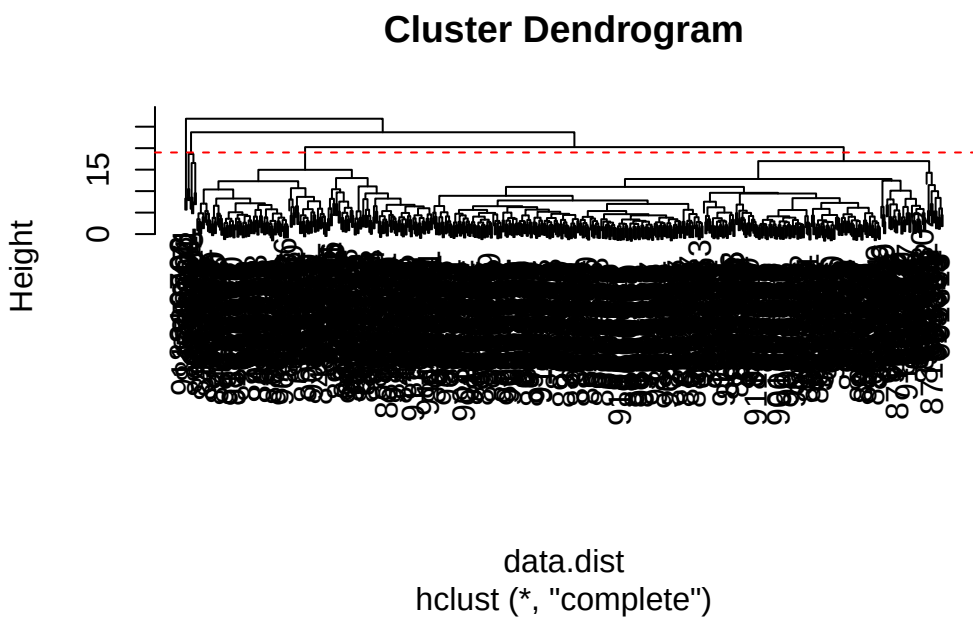
```
          -0.10446933             -0.23663968             -0.22487053
     smoothness_worst       compactness_worst         concavity_worst
          -0.12795256             -0.21009588             -0.22876753
  concave.points_worst          symmetry_worst fractal_dimension_worst
          -0.25088597             -0.12290456             -0.13178394
```

Hierarchical Clustering

The goal of this section is to do hierarchal clustering of the original data. Recall from class that this type of clustering does not assume in advance the number

```r
data.scaled <- scale(wisc.data)
data.dist <- dist(data.scaled)
wisc.hclust <- hclust(data.dist, method = "complete")
```

```r
plot(wisc.hclust)
abline(h = 19, col = "red", lty = 2)
```



Selecting numbers of clusters

```
wisc.hclust.clusters <- cutree(wisc.hclust, k = 4)
```

```
table(wisc.hclust.clusters, diagnosis)
```

```
                    diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
                   2   2   5
                   3 343  40
                   4   0   2
```

Using different methods

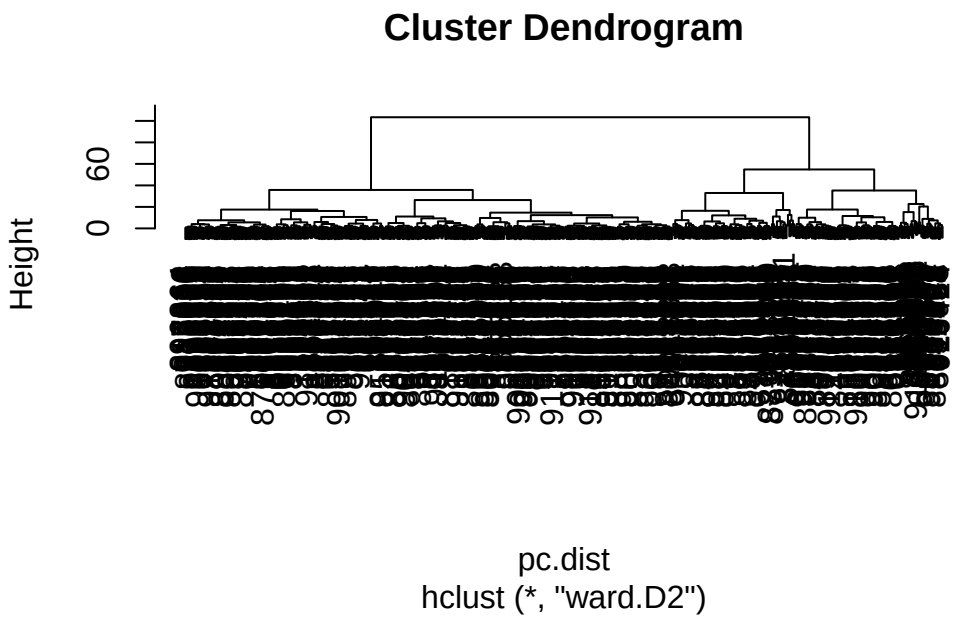
```
wisc.hclust.ward <- hclust(data.dist, method = "ward.D2")
wisc.hclust.clusters.ward <- cutree(wisc.hclust.ward, k = 4)
table(wisc.hclust.clusters.ward, diagnosis)
```

```
                         diagnosis
wisc.hclust.clusters.ward   B   M
                        1   0 115
                        2   6  48
                        3 337  48
                        4  14   1
```

Combining methods

The idea here is that I can take my new variables (i.e. the scores on the PCS `wisc.pr$x`) that are better descriptors of the data-set than the original features (i.e. the 30 columns in `wisc.data`) and use these as a basis for clustering.

```
pc.dist <- dist(wisc.pr$x[,1:3])
wisc.pr.hclust <- hclust(pc.dist, method = "ward.D2")
plot(wisc.pr.hclust)
```




```
grps <- cutree(wisc.pr.hclust, k = 2 )
table(diagnosis)
```

```
diagnosis
  B   M
357 212
```

```
wisc.pr.hclust.clusters <- cutree(wisc.pr.hclust, k=2)
```

I can now run `table()` with both my clustering `grps` and the expert `diagnosis`

> Q13. How well does the newly created hclust model with two clusters separate out the two "M" and "B" diagnoses?

```
table(grps, diagnosis)
```

```
    diagnosis
grps   B   M
   1  24 179
   2 333  33
```

Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the table() function to compare the output of each model (wisc.hclust.clusters and wisc.pr.hclust.clusters) with the vector containing the actual diagnoses.

wisc.hclust.clusters seems to be better in separating the diagnoses.

```
table(wisc.hclust.clusters, diagnosis)
```

```
                    diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
                   2   2   5
                   3 343  40
                   4   0   2
```

```
table(wisc.pr.hclust.clusters, diagnosis)
```

```
                       diagnosis
wisc.pr.hclust.clusters   B   M
                      1  24 179
                      2 333  33
```

Our cluster "1" has 179 "M" diagnosis or cluster "2" has 333 "B" diagnosis

179 TP 24 FP

Sensitivity: TP/(TP + FN)

```
179/(179+33)
```

```
[1] 0.8443396
```

Specificity: TN/(TN+FP)

```
333/(333+24)
```

```
[1] 0.9327731
```

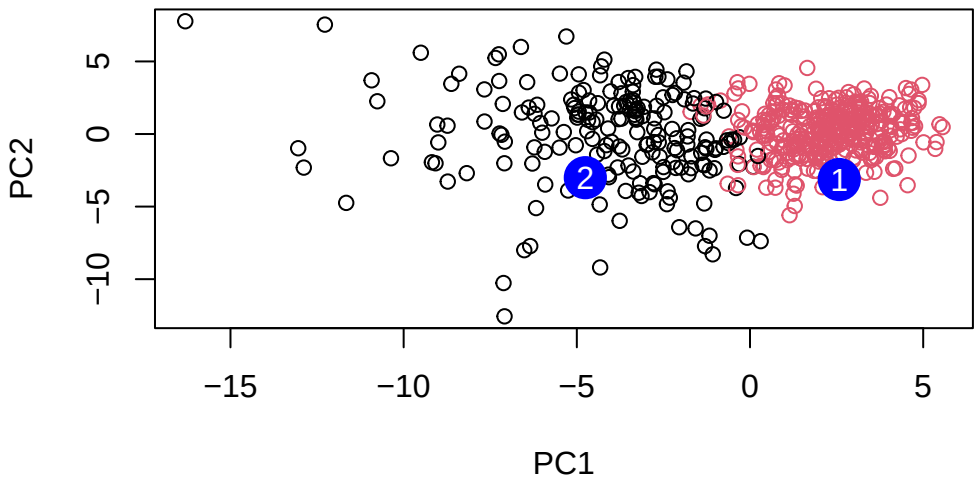
Prediction

We can use our PCA model for predictions of new-unseen cases:

```
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

```
           PC1       PC2        PC3        PC4       PC5        PC6        PC7
[1,]  2.576616 -3.135913  1.3990492 -0.7631950  2.781648 -0.8150185 -0.3959098
[2,] -4.754928 -3.009033 -0.1660946 -0.6052952 -1.140698 -1.2189945  0.8193031
            PC8       PC9       PC10      PC11      PC12      PC13     PC14
[1,] -0.2307350 0.1029569 -0.9272861 0.3411457  0.375921 0.1610764 1.187882
[2,] -0.3307423 0.5281896 -0.4855301 0.7173233 -1.185917 0.5893856 0.303029
          PC15       PC16        PC17        PC18        PC19       PC20
[1,] 0.3216974 -0.1743616 -0.07875393 -0.11207028 -0.08802955 -0.2495216
[2,] 0.1299153  0.1448061 -0.40509706  0.06565549  0.25591230 -0.4289500
           PC21       PC22       PC23       PC24        PC25         PC26
[1,]  0.1228233 0.09358453 0.08347651  0.1223396  0.02124121  0.078884581
[2,] -0.1224776 0.01732146 0.06316631 -0.2338618 -0.20755948 -0.009833238
             PC27        PC28         PC29         PC30
[1,]  0.220199544 -0.02946023 -0.015620933  0.005269029
[2,] -0.001134152  0.09638361  0.002795349 -0.019015820
```

```
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



Q16. Which of these new patients should we prioritize for follow up based on your results?

We should prioritize Patient 2 based on the results.